



## Top 61 CAT Games and Tournamnents Questions With Video Solutions

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# Questions

## Instructions

The game of QUIET is played between two teams. Six teams, numbered 1, 2, 3, 4, 5, and 6, play in a QUIET tournament. These teams are divided equally into two groups. In the tournament, each team plays every other team in the same group only once, and each team in the other group exactly twice. The tournament has several rounds, each of which consists of a few games. Every team plays exactly one game in each round.

The following additional facts are known about the schedule of games in the tournament.

1. Each team played against a team from the other group in Round 8.
2. In Round 4 and Round 7, the match-ups, that is the pair of teams playing against each other, were identical.
3. In Round 5 and Round 8, the match-ups were identical.
3. Team 4 played Team 6 in both Round 1 and Round 2.
4. Team 1 played Team 5 ONLY once and that was in Round 2.
5. Team 3 played Team 4 in Round 3. Team 1 played Team 6 in Round 6.
6. In Round 8, Team 3 played Team 6, while Team 2 played Team 5.

## Question 1

What is the number of the team that played Team 6 in Round 3?

one game in each round.

The following additional facts are known about the schedule of games in the tournament.

1. Each team played against a team from the other group in Round 8.
2. In Round 4 and Round 7, the match-ups, that is the pair of teams playing against each other, were identical. In Round 5 and Round 8, the match-ups were identical.
3. Team 4 played Team 6 in both Round 1 and Round 2.
4. Team 1 played Team 5 ONLY once and that was in Round 2.
5. Team 3 played Team 4 in Round 3. Team 1 played Team 6 in Round 6.
6. In Round 8, Team 3 played Team 6, while Team 2 played Team 5.

Group 1: 2, 3, 4, 5, 6  
Group 2: 1, 2, 3, 4, 5, 6  
 $2 \times 3 \times 3 = 18$   
24

VIDEO SOLUTION

## Question 2

Which team among the teams numbered 2, 3, 4, and 5 was not part of the same group?

- A 5
- B 3
- C 4
- D 2

one game in each round.

The following additional facts are known about the schedule of games in the tournament.

1. Each team played against a team from the other group in Round 8.
2. In Round 4 and Round 7, the match-ups, that is the pair of teams playing against each other, were identical. In Round 5 and Round 8, the match-ups were identical.
3. Team 4 played Team 6 in both Round 1 and Round 2.
4. Team 5 played Team 3 ONLY once and that was in Round 2.
5. Team 3 played Team 4 in Round 3. Team 1 played Team 6 in Round 6.
6. In Round 8, Team 1 played Team 6, while Team 2 played Team 5.

Handwritten notes on a whiteboard:

Group 1:  $2 \times 3 \times 3 = 18$   
 $= 24$

Group 2:  $3 \times 2 = 6$

A man is sitting at a desk with a tablet, pointing at the screen.

VIDEO SOLUTION

### Question 3

How many rounds were there in the tournament?

one game in each round.

The following additional facts are known about the schedule of games in the tournament.

1. Each team played against a team from the other group in Round 8.
2. In Round 4 and Round 7, the match-ups, that is the pair of teams playing against each other, were identical. In Round 5 and Round 8, the match-ups were identical.
3. Team 4 played Team 6 in both Round 1 and Round 2.
4. Team 5 played Team 3 ONLY once and that was in Round 2.
5. Team 3 played Team 4 in Round 3. Team 1 played Team 6 in Round 6.
6. In Round 8, Team 1 played Team 6, while Team 2 played Team 5.

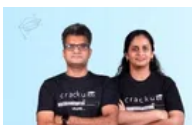
Handwritten notes on a whiteboard:

Group 1:  $2 \times 3 \times 3 = 18$   
 $= 24$

Group 2:  $3 \times 2 = 6$

A man is sitting at a desk with a tablet, pointing at the screen.

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### Question 4

What is the number of the team that played Team 1 in Round 7?

one game in each round.

The following additional facts are known about the schedule of games in the tournament.

1. Each team played against a team from the other group in Round 8.
2. In Round 4 and Round 7, the match-ups, that is the pair of teams playing against each other, were identical. In Round 5 and Round 8, the match-ups were identical.
3. Team 4 played Team 6 in both Round 1 and Round 2.
4. Team 5 played Team 3 ONLY once and that was in Round 2.
5. Team 3 played Team 4 in Round 3. Team 1 played Team 6 in Round 6.
6. In Round 8, Team 1 played Team 6, while Team 2 played Team 5.

Handwritten notes on a whiteboard:

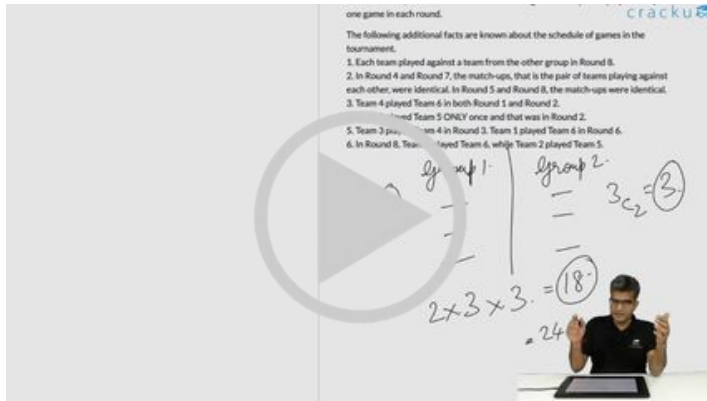
Group 1:  $2 \times 3 \times 3 = 18$   
 $= 24$

Group 2:  $3 \times 2 = 6$

A man is sitting at a desk with a tablet, pointing at the screen.

### Question 5

What is the number of the team that played Team 1 in Round 5?



### Instructions

The following questions relate to a game to be played by you and your friend. The game consists of a 4 x 4 board (see below) where each cell contains a positive integer. You and your friend make moves alternately. A move by any of the players consists of splitting the current board configuration into two equal halves and retaining one of them. In your moves you are allowed to split the board only vertically and to decide to retain either the left or the right half. Your friend, in his/her moves, can split the board only horizontally and can retain either the lower or the upper half. After two moves by each player a single cell will remain which can no longer be split and the number in that cell will be treated as the gain (in rupees) of the person who has started the game. A sample game is shown below. So your gain is Re.1. With the same initial board configuration as above and assuming that you have to make the first move, answer the following questions.

Initial Board

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 2 | 4 |
| 5 | 1 | 6 | 7 |
| 9 | 1 | 3 | 2 |
| 6 | 1 | 8 | 4 |

After your move (retain left)

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | . | . |
| 5 | 1 | . | . |
| 9 | 1 | . | . |
| 6 | 1 | . | . |

After your friends move (retain upper)

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | . | . |
| 5 | 1 | . | . |
| . | . | . | . |
| . | . | . | . |

After your move (retain right)

|   |   |   |   |
|---|---|---|---|
| . | 1 | . | . |
| . | 1 | . | . |
| . | . | . | . |
| . | . | . | . |

After your friends move (retain lower)

|   |   |   |   |
|---|---|---|---|
| . | . | . | . |
|   | 1 |   |   |
|   |   |   |   |
|   |   |   |   |

### Question 6

If both of you select your moves intelligently then at the end of the game your gain will be

- A Rs.4
- B Rs.3
- C Rs.2
- D None of these

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After your friends move (retain upper)

|   |   |   |   |
|---|---|---|---|
| 2 | 1 | 1 | 1 |
| 3 | 1 |   |   |
|   |   |   |   |
|   |   |   |   |

After your move (retain right)

|   |   |  |  |
|---|---|--|--|
| 1 | 1 |  |  |
| 3 |   |  |  |
|   |   |  |  |
|   |   |  |  |

After your friends move (retain lower)

|   |  |  |  |
|---|--|--|--|
| 1 |  |  |  |
|   |  |  |  |
|   |  |  |  |
|   |  |  |  |

If you choose (retain right) (retain left) in your turns, the best move sequence for your friend to reduce your gain to a minimum will be

A) (retain upper)(retain lower) B) (retain lower) (retain upper)  
C) (retain upper)(retain upper) D) (retain lower) (retain lower)

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### Question 7

If your first move is (retain right), then whatever moves your friend may select you can always force a gain of no less than

- A Rs.3
- B Rs.6
- C Rs.4
- D None of these

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After your friends move (retain upper)

|   |   |   |
|---|---|---|
| 2 | 1 | 4 |
| 3 | 1 |   |
|   |   |   |

After your move (retain right)

|   |   |  |
|---|---|--|
| 1 | 1 |  |
|   | 1 |  |
|   |   |  |

After your friends move (retain lower)

|   |   |  |
|---|---|--|
| 1 | 1 |  |
|   | 1 |  |
|   |   |  |

If you choose (retain right) (retain left) in your turns, the best move sequence for your friend to reduce your gain to a minimum will be

A) (retain upper)(retain lower)      B) (retain lower) (retain upper)

C) (retain upper)(retain upper)      D) (retain lower) (retain lower)



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### Question 8

If you choose (retain right) (retain left) in your turns, the best move sequence for your friend to reduce your gain to a minimum will be

- A (retain upper)(retain lower)
- B (retain lower) (retain upper)
- C (retain upper) (retain upper)
- D (retain lower) (retain lower)

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After your friends move (retain upper)

|   |   |   |
|---|---|---|
| 2 | 1 | 4 |
| 3 | 1 |   |
|   |   |   |

After your move (retain right)

|   |   |  |
|---|---|--|
| 1 | 1 |  |
|   | 1 |  |
|   |   |  |

After your friends move (retain lower)

|   |   |  |
|---|---|--|
| 1 | 1 |  |
|   | 1 |  |
|   |   |  |

If you choose (retain right) (retain left) in your turns, the best move sequence for your friend to reduce your gain to a minimum will be

A) (retain upper)(retain lower)      B) (retain lower) (retain upper)

C) (retain upper)(retain upper)      D) (retain lower) (retain lower)



VIDEO SOLUTION

### Instructions

Alia, Badal, Clive, Dilshan, and Ehsaan played a game in which each asks a unique question to all the others and they respond by tapping their feet, either once or twice or thrice. One tap means "Yes", two taps mean "No", and three taps mean "Maybe".

A total of 40 taps were heard across the five questions. Each question received at least one "Yes", one "No", and one "Maybe."

The following information is known.

- Alia tapped a total of 6 times and received 9 taps to her question. She responded "Yes" to the questions asked by both Clive and Dilshan.
- Dilshan and Ehsaan tapped a total of 11 and 9 times respectively. Dilshan responded "No" to Badal.
- Badal, Dilshan, and Ehsaan received equal number of taps to their respective questions.
- No one responded "Yes" more than twice.

5. No one's answer to Alia's question matched the answer that Alia gave to that person's question. This was also true for Ehsaan.

6. Clive tapped more times in total than Badal.

#### Question 9

How many "Yes" responses were received across all the questions?

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#### Question 10

Which two people tapped an equal number of times in total?

- A Badal and Dilshan
- B Clive and Ehsaan
- C Dilshan and Clive
- D Alia and Badal

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#### Question 11

What was Clive's response to Ehsaan's question?

- A No
- B Maybe
- C Cannot be determined
- D Yes

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#### Question 12

How many taps did Clive receive for his question?

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## CAT Syllabus PDF



#### Instructions

Answer the following questions based on the information given below:

In a sports event, six teams (A, B, C, D, E and F) are competing against each other. Matches are scheduled in two stages. Each team plays three matches in Stage – I and two matches in Stage – II. No team plays against the same team more than once in the event. No ties are permitted in any of the matches. The observations after the completion of Stage – I and Stage – II are as given below.

### Stage-I:

- One team won all the three matches.
- Two teams lost all the matches.
- D lost to A but won against C and F.
- E lost to B but won against C and F.
- B lost at least one match.
- F did not play against the top team of Stage-I.

### Stage-II:

- The leader of Stage-I lost the next two matches
- Of the two teams at the bottom after Stage-I, one team won both matches, while the other lost both matches.
- One more team lost both matches in Stage-II.

### Question 13

The team(s) with the most wins in the event is (are):

- A A
- B A & C
- C F
- D E
- E B & E

Stage I

| Name | Won | Lost |
|------|-----|------|
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |

Stage 2

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Answer the following questions based on the information given below.

In a sports event, six teams (A, B, C, D, E and F) are competing against each other. Matches are scheduled in two stages. Each team plays three matches in Stage-I and two matches in Stage-II. No team plays against the same team more than once in the event. No ties are permitted in any of the matches. The observations after the completion of Stage-I and Stage-II are as given below.

Stage-I:

- One team won all the three matches.
- Two teams lost all the matches.
- D lost to A but won against C and F.
- E lost to B but won against C and F.
- B lost at least one match.
- F did not play against the top team of Stage-I.

Stage-II:

- The leader of Stage-I lost the next two matches.
- Of the two teams at the bottom after Stage-I, one team won both matches, while the other lost both matches.
- One more team lost both matches in Stage-II.

VIDEO SOLUTION

### Question 14

The two teams that defeated the leader of Stage-I are:

- A B & F
- B E & F
- C B & D
- D E & D
- E F & D



**Stage 1**

| Name | Won | Lost |
|------|-----|------|
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |

**Stage 2**

|  |
|--|
|  |
|  |
|  |
|  |
|  |

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Answer the following questions based on the information given below:

In a sports event, six teams (A, B, C, D, E and F) are competing against each other. Matches are scheduled in two stages. Each team plays three matches in Stage - I and two matches in Stage - II. No team plays against the same team more than once in the event. No ties are permitted in any of the matches. The observations after the completion of Stage - I and Stage - II are as given below.

**Stage-I:**

- One team won all three matches.
- Two teams lost all the matches.
- Team A but won against C and F.
- Team B won against C and F.
- Team C lost one match.
- Team D did not play against the top team of Stage-I.

**Stage-II:**

- The leader of Stage-I lost the next two matches.
- Two teams at the bottom after Stage-I, one team won both matches, other lost both matches.
- One more team lost both matches in Stage-II.

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### Question 15

The only team(s) that won both matches in Stage-II is (are):

- A B
- B E & F
- C A, E & F
- D B, E & F
- E B & F

**Stage 1**

| Name | Won | Lost |
|------|-----|------|
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |
|      |     |      |

**Stage 2**

|  |
|--|
|  |
|  |
|  |
|  |
|  |

**cracku**

Answer the following questions based on the information given below:

In a sports event, six teams (A, B, C, D, E and F) are competing against each other. Matches are scheduled in two stages. Each team plays three matches in Stage - I and two matches in Stage - II. No team plays against the same team more than once in the event. No ties are permitted in any of the matches. The observations after the completion of Stage - I and Stage - II are as given below.

**Stage-I:**

- One team won all three matches.
- Two teams lost all the matches.
- Team A but won against C and F.
- Team B won against C and F.
- Team C lost one match.
- Team D did not play against the top team of Stage-I.

**Stage-II:**

- The leader of Stage-I lost the next two matches.
- Two teams at the bottom after Stage-I, one team won both matches, other lost both matches.
- One more team lost both matches in Stage-II.

VIDEO SOLUTION

# CAT Formula PDF

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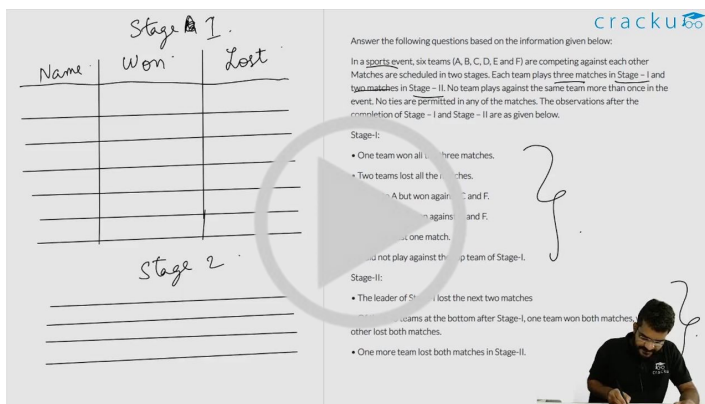
### Question 16

The teams that won exactly two matches in the event are:

- A A, D & F
- B D & E
- C E & F

D D, E & F

E D & F



VIDEO SOLUTION

### Instructions

Directions for the following four questions: Answer the questions on the basis of the information given below.

The year was 2006. All six teams in Pool A of World Cup hockey, play each other exactly once.

Each win earns a team three points, a draw earns one point and a loss earns zero points. The two teams with the highest points qualify for the semifinals. In case of a tie, the team with the highest goal difference (Goal For . Goals Against) qualifies.

In the opening match, Spain lost to Germany.

After the second round (after each team played two matches), the pool table looked as shown below.

| Teams        | Games Played | Won | Drawn | Lost | Goals For | Goals Against | Points |
|--------------|--------------|-----|-------|------|-----------|---------------|--------|
| Germany      | 2            | 2   | 0     | 0    | 3         | 1             | 6      |
| Argentina    | 2            | 2   | 0     | 0    | 2         | 0             | 6      |
| Spain        | 2            | 1   | 0     | 1    | 5         | 2             | 3      |
| Pakistan     | 2            | 1   | 0     | 1    | 2         | 1             | 3      |
| New Zealand  | 2            | 0   | 0     | 2    | 1         | 6             | 0      |
| South Africa | 2            | 0   | 0     | 2    | 1         | 4             | 0      |

In the third round, Spain played Pakistan, Argentina played Germany, and New Zealand played South Africa. All the third round matches were drawn. The following are some results from the fourth and fifth round matches

- (a) Spain won both the fourth and fifth round matches.
- (b) Both Argentina and Germany won their fifth round matches by 3 goals to 0.
- (c) Pakistan won both the fourth and fifth round matches by 1 goal to 0.

### Question 17

Which one of the following statements is true about matches played in the first two rounds?

- A Germany beat New Zealand by 1 goal to 0.
- B Spain beat New Zealand by 4 goals to 0.
- C Spain beat South Africa by 2 goals to 0.
- D Germany beat South Africa by 2 goals to 1.

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### Question 18

Which one of the following statements is true about matches played in the first two rounds?

- A Pakistan beat South Africa by 2 goals to 1.
- B Argentina beat Pakistan by 1 goal to 0.
- C Germany beat Pakistan by 2 goals to 1.
- D Germany beat Spain by 2 goals to 1.

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### Question 19

If Pakistan qualified as one of the two teams from Pool A, which was the other team that qualified?

- A Argentina
- B Germany
- C Spain
- D Cannot be determined

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### Question 20

Which team finished at the top of the pool after five rounds of matches?

- A Argentina
- B Germany
- C Spain
- D Cannot be determined

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### Instructions

In the table below is the listing of players, seeded from highest (#1) to lowest (#32), who are due to play in an Association of Tennis Players (ATP) tournament for women. This tournament has four knockout rounds before the final, i.e., first round, second round, quarterfinals, and semi-finals. In the first round, the highest seeded player plays the lowest seeded player (seed # 32) which is designated match No. 1 of first round; the 2nd seeded player plays the 31st seeded player which is designated match No. 2 of the first round, and so on. Thus, for instance, match No. 16 of first round is to be played between 16th seeded player and the 17th seeded player. In the second round, the winner of match No. 1 of first round plays the winner of match No. 16 of first round and is designated match No. 1 of second round. Similarly, the winner of match No. 2 of first round plays the winner of match No. 15 of first round, and is designated match No. 2 of second round. Thus, for instance, match No. 8 of the second round is to be played between the winner of match No. 8 of first round and the winner of match No. 9 of first round. The same pattern is followed for later rounds as well.

| Seed # | Name of Player       |
|--------|----------------------|
| 1.     | Maria Sharapova      |
| 2.     | Lindsay Davenport    |
| 3.     | Amelie Mauresmo      |
| 4.     | Kim Clijsters        |
| 5.     | Svetana Kuznetsova   |
| 6.     | Elena Dementieva     |
| 7.     | Justine Henin        |
| 8.     | Serena Williams      |
| 9.     | Nadia Petrova        |
| 10.    | Venus Williams       |
| 11.    | Patty Schnyder       |
| 12.    | Mary Pierce          |
| 13.    | Anastasia Myskina    |
| 14.    | Alicia Molik         |
| 15.    | Nathalie Dechy       |
| 16.    | Elena Bovina         |
| 17.    | Jelena Jankovic      |
| 18.    | Ana Ivanovic         |
| 19.    | Vera Zvonareva       |
| 20.    | Elena Likhovtseva    |
| 21.    | Daniela Hantuchova   |
| 22.    | Dinara Safina        |
| 23.    | Silvia Farina Elia   |
| 24.    | Tatiana Golovin      |
| 25.    | Shinobu Asagoe       |
| 26.    | Francesca Schiavone  |
| 27.    | Nicole Pietrangeli   |
| 28.    | Gila Dulko           |
| 29.    | Flavia Pennetta      |
| 30.    | Anna Chakvetadze     |
| 31.    | Ai Sugiyama          |
| 32.    | Anna-Lena Groenefeld |

### Question 21

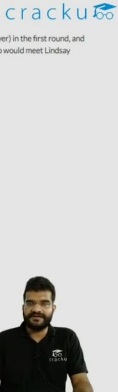
If there are no upsets (a lower seeded player beating a higher seeded player) in the first round, and only match Nos. 6, 7, and 8 of the second round result in upsets, then who would meet Lindsay Davenport in quarter finals, in case Davenport reaches quarter finals?

- A Justine Henin
- B Nadia Petrova
- C Patty Schnyder
- D Venus Williams

In the table below is the listing of players, seeded from highest (#1) to lowest (#32), who are due to play in an Association of Tennis Players (ATP) tournament for women. This tournament has four knockout rounds before the final, i.e., first round, second round, quarterfinals, and semi-finals. In the first round, the highest seeded player plays the lowest seeded player (seed # 32) which is designated match No. 1 of first round; the 2nd seeded player plays the 31st seeded player which is designated match No. 2 of the first round, and so on. Thus, for instance, match No. 16 of first round is to be played between 16th seeded player and the 17th seeded player. In the second round, the winner of match No. 1 of first round plays the winner of match No. 16 of first round and is designated match No. 1 of second round. Similarly, the winner of match No. 2 of first round plays the winner of match No. 15 of first round, and is designated match No. 2 of second round. Thus, for instance, match No. 8 of the second round is to be played between the winner of match No. 8 of first round and the winner of match No. 9 of first round. The same pattern is followed for later rounds as well.

Question (1)

If there are no upsets (a lower seeded player beating a higher seeded player) in the first round, and only match Nos. 6, 7, and 8 of the second round result in upsets, then who would meet Lindsay Davenport in quarter finals, in case Davenport reaches quarter finals?



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### Question 22

If, in the first round, all even numbered matches (and none of the odd numbered ones) result in upsets, and there are no upsets in the second round, then who could be the lowest seeded player facing Maria Sharapova in semi-finals?

- A Anastasia Myskina
- B Flavia Pennetta
- C Nadia Petrova
- D Svetlana Kuznetsova

In the table below is the listing of players, seeded from highest (#1) to lowest (#32), who are due to play in an Association of Tennis Players (ATP) tournament for women. This tournament has four knockout rounds before the final, i.e., first round, second round, quarterfinals, and semi-finals. In the first round, the highest seeded player plays the lowest seeded player (seed # 32) which is designated match No. 1 of first round; the 2nd seeded player plays the 31st seeded player which is designated match No. 2 of the first round, and so on. Thus, for instance, match No. 16 of first round is to be played between 16th seeded player and the 17th seeded player. In the second round, the winner of match No. 1 of first round plays the winner of match No. 16 of first round and is designated match No. 1 of second round. Similarly, the winner of match No. 2 of first round plays the winner of match No. 15 of first round, and is designated match No. 2 of second round. Thus, for instance, match No. 8 of the second round is to be played between the winner of match No. 8 of first round and the winner of match No. 9 of first round. The same pattern is followed for later rounds as well.

| Seed | Name of Player      |
|------|---------------------|
| 1    | Marion Bartoli      |
| 2    | Francesca Schiavone |
| 3    | Anna-Lena Freixas   |
| 4    | Flavia Pennetta     |
| 5    | Kim Clijsters       |
| 6    | Amelie Mauresmo     |
| 7    | Elena Dementieva    |
| 8    | Lindsay Davenport   |
| 9    | Nadia Petrova       |
| 10   | Svetlana Kuznetsova |
| 11   | Alona Bondarenko    |
| 12   | Alisa Kleybanova    |
| 13   | Yanina Knapkova     |
| 14   | Yulia Fedak         |
| 15   | Yulia Glushko       |
| 16   | Yulia Glushko       |
| 17   | Yulia Glushko       |
| 18   | Yulia Glushko       |
| 19   | Yulia Glushko       |
| 20   | Yulia Glushko       |
| 21   | Yulia Glushko       |
| 22   | Yulia Glushko       |
| 23   | Yulia Glushko       |
| 24   | Yulia Glushko       |
| 25   | Yulia Glushko       |
| 26   | Yulia Glushko       |
| 27   | Yulia Glushko       |
| 28   | Yulia Glushko       |
| 29   | Yulia Glushko       |
| 30   | Yulia Glushko       |
| 31   | Yulia Glushko       |
| 32   | Yulia Glushko       |

Question (1)

If there are no upsets (a lower seeded player beating a higher seeded player) in the first round, and only match Nos. 6, 7, and 8 of the second round result in upsets, then who would meet Lindsay Davenport in quarter finals, in case Davenport reaches quarter finals?

#### VIDEO SOLUTION

### Question 23

If the top eight seeds make it to the quarterfinals, then who, amongst the players listed below, would definitely not play against Maria Sharapova in the final, in case Sharapova reaches the final?

- A Amelie Mauresmo
- B Elena Dementieva
- C Kim Clijsters
- D Lindsay Davenport

In the table below is the listing of players, seeded from highest (#1) to lowest (#32), who are due to play in an Association of Tennis Players (ATP) tournament for women. This tournament has four knockout rounds before the final, i.e., first round, second round, quarterfinals, and semi-finals. In the first round, the highest seeded player plays the lowest seeded player (seed # 32) which is designated match No. 1 of first round; the 2nd seeded player plays the 31st seeded player which is designated match No. 2 of the first round, and so on. Thus, for instance, match No. 16 of first round is to be played between 16th seeded player and the 17th seeded player. In the second round, the winner of match No. 1 of first round plays the winner of match No. 16 of first round and is designated match No. 1 of second round. Similarly, the winner of match No. 2 of first round plays the winner of match No. 15 of first round, and is designated match No. 2 of second round. Thus, for instance, match No. 8 of the second round is to be played between the winner of match No. 8 of first round and the winner of match No. 9 of first round. The same pattern is followed for later rounds as well.

| Seed | Name of Player      |
|------|---------------------|
| 1    | Marion Bartoli      |
| 2    | Francesca Schiavone |
| 3    | Anna-Lena Freixas   |
| 4    | Flavia Pennetta     |
| 5    | Kim Clijsters       |
| 6    | Amelie Mauresmo     |
| 7    | Elena Dementieva    |
| 8    | Lindsay Davenport   |
| 9    | Nadia Petrova       |
| 10   | Svetlana Kuznetsova |
| 11   | Alona Bondarenko    |
| 12   | Alisa Kleybanova    |
| 13   | Yanina Knapkova     |
| 14   | Yulia Fedak         |
| 15   | Yulia Glushko       |
| 16   | Yulia Glushko       |
| 17   | Yulia Glushko       |
| 18   | Yulia Glushko       |
| 19   | Yulia Glushko       |
| 20   | Yulia Glushko       |
| 21   | Yulia Glushko       |
| 22   | Yulia Glushko       |
| 23   | Yulia Glushko       |
| 24   | Yulia Glushko       |
| 25   | Yulia Glushko       |
| 26   | Yulia Glushko       |
| 27   | Yulia Glushko       |
| 28   | Yulia Glushko       |
| 29   | Yulia Glushko       |
| 30   | Yulia Glushko       |
| 31   | Yulia Glushko       |
| 32   | Yulia Glushko       |

Question (1)

If there are no upsets (a lower seeded player beating a higher seeded player) in the first round, and only match Nos. 6, 7, and 8 of the second round result in upsets, then who would meet Lindsay Davenport in quarter finals, in case Davenport reaches quarter finals?

#### VIDEO SOLUTION

### Question 24

If Elena Dementieva and Serena Williams lose in the second round, while Justine Henin and Nadia Petrova make it to the semi-finals, then who would play Maria Sharapova in the quarterfinals, in the event Sharapova reaches quarterfinals?

- A Dinara Safina
- B Justine Henin
- C Nadia Petrova
- D Patty Schnyder

In the table below is the listing of players, seeded from highest (#1) to lowest (#32), who are due to play in an Association of Tennis Players (ATP) tournament for women. This tournament has four knockout rounds before the final, i.e., first round, second round, quarterfinals, and semi-finals. In the first round, the highest seeded player plays the lowest seeded player (seed #32) which is designated match No. 1 of first round; the 2nd seeded player plays the 31st seeded player which is designated match No. 2 of the first round, and so on. Thus, for instance, match No. 16 of first round is to be played between 16th seeded player and the 17th seeded player. In the second round, the winner of match No. 1 of first round plays the winner of match No. 16 of first round and is designated match No. 1 of second round. Similarly, the winner of match No. 2 of first round plays the winner of match No. 15 of first round, and is designated match No. 2 of second round. Thus, for instance, match No. 8 of the second round is to be played between the winner of match No. 8 of first round and the winner of match No. 9 of first round. The same pattern is followed for later rounds as well.

| Seed | Player              |
|------|---------------------|
| 1    | Elena Dementieva    |
| 2    | Serena Williams     |
| 3    | Justine Henin       |
| 4    | Nadia Petrova       |
| 5    | Dinara Safina       |
| 6    | Patty Schnyder      |
| 7    | Maria Sharapova     |
| 8    | Francesca Schiavone |
| 9    | Kim Clijsters       |
| 10   | Alona Bondarenko    |
| 11   | Yanina Knapkova     |
| 12   | Lucy Karsch         |
| 13   | Anna-Lena Freixas   |
| 14   | Alisa Kleybanova    |
| 15   | Marion Bartoli      |
| 16   | Francesca Schiavone |
| 17   | Lucy Karsch         |
| 18   | Anna-Lena Freixas   |
| 19   | Alisa Kleybanova    |
| 20   | Marion Bartoli      |
| 21   | Francesca Schiavone |
| 22   | Lucy Karsch         |
| 23   | Anna-Lena Freixas   |
| 24   | Alisa Kleybanova    |
| 25   | Marion Bartoli      |
| 26   | Francesca Schiavone |
| 27   | Lucy Karsch         |
| 28   | Anna-Lena Freixas   |
| 29   | Alisa Kleybanova    |
| 30   | Marion Bartoli      |
| 31   | Francesca Schiavone |
| 32   | Lucy Karsch         |

**Question (1)**

If there are no upsets (a lower seeded player beating a higher seeded player) in the first round, and only match Nos. 6, 7, and 8 of the second round result in upsets, then who would meet Lindsay Davenport in quarter finals, in case Davenport reaches quarter finals?

VIDEO SOLUTION

## CAT Score Calculator

### Instructions

10 players - P1, P2, ... , P10 - competed in an international javelin throw event. The number (after P) of a player reflects his rank at the beginning of the event, with rank 1 going to the topmost player. There were two phases in the event with the first phase consisting of rounds 1, 2, and 3, and the second phase consisting of rounds 4, 5, and 6. A throw is measured in terms of the distance it covers (in meters, up to one decimal point accuracy), only if the throw is a 'valid' one. For an invalid throw, the distance is taken as zero. A player's score at the end of a round is the maximum distance of all his throws up to that round. Players are re-ranked after every round based on their current scores. In case of a tie in scores, the player with a prevailing higher rank retains the higher rank. This ranking determines the order in which the players go for their throws in the next round.

In each of the rounds in the first phase, the players throw in increasing order of their latest rank, i.e. the player ranked 1 at that point throws first, followed by the player ranked 2 at that point and so on. The top six players at the end of the first phase qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance(in m) |
|--------|----------------|
| P1     | 82.9           |
| P3     | 81.5           |
| P5     | 86.4           |
| P6     | 82.5           |
| P7     | 87.2           |
| P9     | 84.1           |

Distances covered by all the valid throws in the third round

| Player | Distance(in m) |
|--------|----------------|
| P1     | 88.6           |
| P3     | 79.0           |
| P9     | 81.4           |

The following facts are also known.

- Among the throws in the second round, only the last two were valid. Both the throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the players in the current round) to throw in the previous round, then the player is said to get a double. Two players got a double.
- In each round of the second phase, exactly one player improved his score. Each of these improvements was by the same amount.
- The gold and bronze medalists improved their scores in the fifth and the sixth rounds respectively. One medal winner improved his score in the fourth round.
- The difference between the final scores of the gold medalist and the silver medalist, as well as the difference between the final scores of the silver medalist and the bronze medalist was 1.0 m.

### Question 25

Which two players got the double?

- P1, P8
- P2, P4
- P8, P10
- P1, P10

cfedk

1 2 3 4 5 6 7 8 9 10

P1 P2 P3 P4 P5 P6 P7 P8 P9 P10

Order

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance (m) |
|--------|--------------|
| P1     | 82.9         |
| P3     | 81.5         |
| P5     | 86.4         |
| P6     | 82.5         |
| P7     | 87.2         |
| P9     | 84.1         |

Distances covered by all the valid throws in the third round

| Player | Distance (m) |
|--------|--------------|
| P1     | 88.6         |
| P3     | 79.0         |
| P9     | 81.4         |

The following facts are also known.

- Among the throws in the second round, only the last two were valid. Both the throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the player is said to get a double. Two players got a double.
- In each round of the second phase, exactly one player improved his score. Each of these improvements was by the same amount.
- The gold and bronze medalists improved their scores in the fifth and the sixth rounds respectively. One medal winner improved his score in the fourth round.
- The difference between the final scores of the gold medalist and the silver medalist, as well as the difference between the final scores of the silver medalist and the bronze medalist was 1.0 m.

VIDEO SOLUTION

### Question 26

Who won the silver medal?



- A P5
- B P7
- C P9
- D P1

Cracku logo and handwritten notes on the left side of the image.

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance (m) |
|--------|--------------|
| P1     | 82.5         |
| P3     | 81.5         |
|        | 86.4         |
|        | 82.5         |
|        | 87.2         |
|        | 81.1         |

Distances covered by all the valid throws in the third round

| Player | Distance (m) |
|--------|--------------|
| P1     | 88.6         |
| P3     | 79           |
| P9     | 81.4         |

The following are also known:

- i. Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- ii. If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

### Question 27

By how much did the gold medalist improve his score (in m) in the second phase?

- A 1.0
- B 2.0
- C 2.4
- D 1.2

Cracku logo and handwritten notes on the left side of the image.

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance (m) |
|--------|--------------|
| P1     | 82.5         |
| P3     | 81.5         |
|        | 86.4         |
|        | 82.5         |
|        | 87.2         |
|        | 81.1         |

Distances covered by all the valid throws in the third round

| Player | Distance (m) |
|--------|--------------|
| P1     | 88.6         |
| P3     | 79           |
| P9     | 81.4         |

The following are also known:

- i. Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- ii. If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

Cracku logo and image of two people on the left side of the banner.

# CAT Crash Course

Live Classes by IIMA alumni

Cracku logo and image of a person on the right side of the banner.

### Question 28

Which of the following can be the final score (in m) of P8?



- A 81.9
- B 0
- C 82.7
- D 85.1

Handwritten notes on the left: Order  
 1. P1  
 2. P7  
 3. P5  
 4. P9  
 5. P1  
 6. P6  
 7. P3  
 8. P7  
 9. P9  
 10. P10

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance (m) |
|--------|--------------|
| P1     | 82.5         |
| P2     | 81.5         |
| P3     | 86.4         |
| P4     | 82.5         |
| P5     | 87.2         |
| P6     | 81.1         |

Distances covered by all the valid throws in the third round

| Player | Distance (m) |
|--------|--------------|
| P1     | 88.6         |
| P3     | 79           |
| P9     | 81.4         |

The following facts are also known:

- i. Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- ii. If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

### Question 29

Who threw the last javelin in the event?

- A P7
- B P1
- C P9
- D P10

Handwritten notes on the left: Order  
 1. P1  
 2. P7  
 3. P5  
 4. P9  
 5. P1  
 6. P6  
 7. P3  
 8. P7  
 9. P9  
 10. P10

qualify for the second phase. In each of the rounds in the second phase, the players throw in decreasing order of their latest rank i.e. the player ranked 6 at that point throws first, followed by the player ranked 5 at that point and so on. The players ranked 1, 2, and 3 at the end of the sixth round receive gold, silver, and bronze medals respectively.

All the valid throws of the event were of distinct distances (as per stated measurement accuracy). The tables below show distances (in meters) covered by all valid throws in the first and the third round in the event.

Distances covered by all the valid throws in the first round

| Player | Distance (m) |
|--------|--------------|
| P1     | 82.5         |
| P2     | 81.5         |
| P3     | 86.4         |
| P4     | 82.5         |
| P5     | 87.2         |
| P6     | 81.1         |

Distances covered by all the valid throws in the third round

| Player | Distance (m) |
|--------|--------------|
| P1     | 88.6         |
| P3     | 79           |
| P9     | 81.4         |

The following facts are also known:

- i. Among the throws in the second round, only the last two were valid. Both throws enabled these players to qualify for the second phase, with one of them qualifying with the least score. None of these players won any medal.
- ii. If a player throws first in a round AND he was also the last (among the current round) to throw in the previous round, then the player gets a double. Two players got a double.

VIDEO SOLUTION

### Question 30

What was the final score (in m) of the silver-medalist?

- A 89.6
- B 88.4

C 88.6

D 87.2

Cracku logo and handwritten notes on the left margin: P1, P2, P3, P4, P5, P6, P7, P8, P9, P10. The main content area contains a table of distances in meters for the first and third rounds of a throw event. The first round table lists distances for P1 (82.1), P3 (81.5), P5 (80.4), P7 (82.4), P9 (87.2), and P10 (81.1). The third round table lists distances for P1 (88.6), P3 (79), and P9 (81.4). The text explains that the top four teams from each group advance to the second stage, while the rest are eliminated. The second stage involves several rounds, with the winner of a match advancing to the next round and the loser being eliminated. The team that remains undefeated in the second stage is declared the winner and claims the Gold Cup. The tournament rules state that each match results in a winner and a loser with no possibility of a tie. In the first stage, a team earns one point for each win and no points for a loss. At the end of the first stage, teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

VIDEO SOLUTION

5000+ CAT Questions

Cracku CAT Questions

Cracku logo and a 3D character holding a question mark.

## Instructions

Directions for the next 5 questions:

Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament is conducted in two stages. In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds. A round involves one match for each team. The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner and claims the Gold Cup.

The tournament rules are such that each match results in a winner and a loser with no possibility of a tie. In the first stage a team earns one point for each win and no points for a loss. At the end of the first stage teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

## Question 31

What is the highest number of wins for a team in the first stage in spite of which it would be eliminated at the end of first stage?

A 1

B 2

C 3

D 5

cracku

Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament is conducted in two stages. In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds. A round involves one match for each team. The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the champion and claims the Gold Cup.

The tournament is such that each match results in a winner and a loser with no possibility of a tie. In the first stage a team earns one point for each win and no points for a loss. At the end of the first stage teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

Handwritten notes:  $8C_2 \times 2 \Rightarrow 56$ ,  $8/4$ ,  $W \rightarrow 1$ ,  $L \rightarrow 0$

VIDEO SOLUTION

### Question 32

What is the total number of matches played in the tournament?

- A 28
- B 55
- C 63
- D 35

cracku

Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament is conducted in two stages. In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds. A round involves one match for each team. The winner of a match in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the champion and claims the Gold Cup.

The tournament is such that each match results in a winner and a loser with no possibility of a tie. In the first stage a team earns one point for each win and no points for a loss. At the end of the first stage teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

Handwritten notes:  $8C_2 \times 2 \Rightarrow 56$ ,  $8/4$ ,  $W \rightarrow 1$ ,  $L \rightarrow 0$

VIDEO SOLUTION

### Question 33

The minimum number of wins needed for a team in the first stage to guarantee its advancement to the next stage is:

- A 5
- B 6
- C 7
- D 4

cracku

Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament is conducted in two stages. In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds. A round involves one match for each team. The winning team in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner and claims the Gold Cup.

The tournament is such that each match results in a winner and a loser with no possibility of a tie. In the first stage a team earns one point for each win and no points for a loss. At the end of the first stage teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

Handwritten notes:  $8C_2 \times 2 \Rightarrow 56$ ,  $8/4 \Rightarrow 2$ ,  $W \rightarrow 1$ ,  $L \rightarrow 0$

VIDEO SOLUTION

cracku

# CAT Daily Targets

Step by step, heading towards 99+ percentile

## Question 34

What is the number of rounds in the second stage of the tournament?

- A 1
- B 2
- C 3
- D 4

cracku

Sixteen teams have been invited to participate in the ABC Gold Cup cricket tournament. The tournament is conducted in two stages. In the first stage, the teams are divided into two groups. Each group consists of eight teams, with each team playing every other team in its group exactly once. At the end of the first stage, the top four teams from each group advance to the second stage while the rest are eliminated. The second stage comprises of several rounds. A round involves one match for each team. The winning team in a round advances to the next round, while the loser is eliminated. The team that remains undefeated in the second stage is declared the winner and claims the Gold Cup.

The tournament is such that each match results in a winner and a loser with no possibility of a tie. In the first stage a team earns one point for each win and no points for a loss. At the end of the first stage teams in each group are ranked on the basis of total points to determine the qualifiers advancing to the next stage. Ties are resolved by a series of complex tie-breaking rules so that exactly four teams from each group advance to the next stage.

Handwritten notes:  $8C_2 \times 2 \Rightarrow 56$ ,  $8/4 \Rightarrow 2$ ,  $W \rightarrow 1$ ,  $L \rightarrow 0$

VIDEO SOLUTION

## Question 35

Which of the following statements is true?

- A The winner will have more wins than any other team in the tournament.
- B At the end of the first stage, no team eliminated from the tournament will have more wins than any of the teams qualifying for the second stage.

- C** It is possible that the winner will have the same number of wins in the entire tournament as a team eliminated at the end of the first stage.
- D** The number of teams with exactly one win in the second stage of the tournament is 4.

[VIEW SOLUTION](#)

### Instructions

The following facts are known about the goals scored by these four players only. All the questions refer only to the goals scored by these four players.

The management of a university hockey team was evaluating performance of four women players - Amla, Bimla, Harita and Sarita for their possible selection in the university team for next year. For this purpose, the management was looking at the number of goals scored by them in the past 8 matches, numbered 1 through 8. The four players together had scored a total of 12 goals in these matches. In the 8 matches, each of them had scored at least one goal. No two players had scored the same total number of goals.

1. Only one goal was scored in every even numbered match.
2. Harita scored more goals than Bimla.
3. The highest goal scorer scored goals in exactly 3 matches including Match 4 and Match 8.
4. Bimla scored a goal in Match 1 and one each in three other consecutive matches.
5. An equal number of goals were scored in Match 3 and Match 7, which was different from the number of goals scored in either Match 1 or Match 5.
6. The match in which the highest number of goals was scored was unique and it was not Match 5.

### Question 36

How many goals were scored in Match 7?

- A** 3
- B** 2
- C** 1
- D** Cannot be determined

[VIDEO SOLUTION](#)

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## CAT Sectional Tests

**Take Here**

### Question 37

Which of the following is the correct sequence of goals scored in matches 1, 3, 5 and 7?

- A 5, 1, 0, 1
- B 3, 1, 2, 1
- C 3, 2, 1, 2
- D 4, 1, 2, 1

The management of a university hockey team was evaluating performance of four women players - Amla, Binla, Harita and Sarita for their possible selection in the university team for next year. For this purpose, the management was looking at the number of goals scored by them in the past 8 matches, numbered 1 through 8. The four players together had scored a total of 12 goals in these matches. In the 8 matches, each of them had scored at least one goal. No two players had scored the same total number of goals.

1. Amla was scored in every even numbered match.  
 2. Harita scored more goals than Binla.  
 3. The highest goal scorer scored goals in exactly 3 matches including Match 4 and Match 8.  
 4. Binla scored a goal in Match 1 and one each in three other consecutive matches.  
 5. The total number of goals scored were scored in Match 3 and Match 7, which was different from the number of goals scored in either Match 1 or Match 5.  
 6. The number of goals scored in Match 2 and Match 6 was unique and it was different from the number of goals scored in any other match.

|   | A | B | H | S |
|---|---|---|---|---|
| 1 |   |   |   |   |
| 2 |   |   |   |   |
| 3 |   |   |   |   |
| 4 |   |   |   |   |
| 5 |   |   |   |   |
| 6 |   |   |   |   |
| 7 |   |   |   |   |
| 8 |   |   |   |   |

Handwritten calculations:  
 $1+2+3+4=10$   
 $2+3+4+5=14$   
 3

VIDEO SOLUTION

### Question 38

Which of the following statement(s) is/are true?

Statement-1: Amla and Sarita never scored goals in the same match.

Statement-2: Harita and Sarita never scored goals in the same match.

- A Statement-1 only
- B Statement-2 only
- C Both the statements
- D None of the statements

The management of a university hockey team was evaluating performance of four women players - Amla, Binla, Harita and Sarita for their possible selection in the university team for next year. For this purpose, the management was looking at the number of goals scored by them in the past 8 matches, numbered 1 through 8. The four players together had scored a total of 12 goals in these matches. In the 8 matches, each of them had scored at least one goal. No two players had scored the same total number of goals.

1. Amla was scored in every even numbered match.  
 2. Harita scored more goals than Binla.  
 3. The highest goal scorer scored goals in exactly 3 matches including Match 4 and Match 8.  
 4. Binla scored a goal in Match 1 and one each in three other consecutive matches.  
 5. The total number of goals scored were scored in Match 3 and Match 7, which was different from the number of goals scored in either Match 1 or Match 5.  
 6. The number of goals scored in Match 2 and Match 6 was unique and it was different from the number of goals scored in any other match.

|   | A | B | H | S |
|---|---|---|---|---|
| 1 |   |   |   |   |
| 2 |   |   |   |   |
| 3 |   |   |   |   |
| 4 |   |   |   |   |
| 5 |   |   |   |   |
| 6 |   |   |   |   |
| 7 |   |   |   |   |
| 8 |   |   |   |   |

Handwritten calculations:  
 $1+2+3+4=10$   
 $2+3+4+5=14$   
 3

VIDEO SOLUTION

### Question 39

Which of the following statement(s) is/are false?

Statement-1: In every match at least one player scored a goal.

Statement-2: No two players scored goals in the same number of matches.



- A None of the statements
- B Statement-1 only
- C Both the statements
- D Statement-2 only

cracku

|   | A | B | H | S |
|---|---|---|---|---|
| 1 |   |   |   |   |
| 2 |   |   |   |   |
| 3 |   |   |   |   |
| 4 |   |   |   |   |
| 5 |   |   |   |   |
| 6 |   |   |   |   |
| 7 |   |   |   |   |
| 8 |   |   |   |   |

The management of a university hockey team was evaluating performance of four women players - Amla, Binla, Harita and Sarita for their possible selection in the university team for next year. For this purpose, the management was looking at the number of goals scored by them in the past 8 matches, numbered 1 through 8. The four players together had scored a total of 12 goals in these matches. In the 8 matches, each of them had scored at least one goal. No two players had scored the same total number of goals.

1. Amla was scored in every even numbered match.
2. Harita scored more goals than Binla.
3. The highest goal scorer scored goals in exactly 3 matches including Match 4 and Match 8.
4. Binla scored a goal in Match 1 and one each in three other consecutive matches.

Additional information: The number of goals scored in Match 3 and Match 7, which was different, was the same as the number of goals scored in either Match 1 or Match 5. The highest number of goals scored was unique and it was 4.

Handwritten calculations:

$$1+2+3+4=10$$

$$2+3+4+5=14$$

Handwritten circled numbers: 1, 2, 3, 4, 5, 6, 7, 8

VIDEO SOLUTION

cracku

# CAT Study Material

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## Question 40

If Harita scored goals in one more match as compared to Sarita, which of the following statement(s) is/are necessarily true?

Statement-1: Amla scored goals in consecutive matches.

Statement-2: Sarita scored goals in consecutive matches.

- A Statement-2 only
- B None of the statements
- C Statement-1 only
- D Both the statements

cracku

|   | A | B | H | S |
|---|---|---|---|---|
| 1 |   |   |   |   |
| 2 |   |   |   |   |
| 3 |   |   |   |   |
| 4 |   |   |   |   |
| 5 |   |   |   |   |
| 6 |   |   |   |   |
| 7 |   |   |   |   |
| 8 |   |   |   |   |

The management of a university hockey team was evaluating performance of four women players - Amla, Binla, Harita and Sarita for their possible selection in the university team for next year. For this purpose, the management was looking at the number of goals scored by them in the past 8 matches, numbered 1 through 8. The four players together had scored a total of 12 goals in these matches. In the 8 matches, each of them had scored at least one goal. No two players had scored the same total number of goals.

1. Amla was scored in every even numbered match.
2. Harita scored more goals than Binla.
3. The highest goal scorer scored goals in exactly 3 matches including Match 4 and Match 8.
4. Binla scored a goal in Match 1 and one each in three other consecutive matches.

Additional information: The number of goals scored in Match 3 and Match 7, which was different, was the same as the number of goals scored in either Match 1 or Match 5. The highest number of goals scored was unique and it was 4.

Handwritten calculations:

$$1+2+3+4=10$$

$$2+3+4+5=14$$

Handwritten circled numbers: 1, 2, 3, 4, 5, 6, 7, 8

## Instructions

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak played the Hi-Lo game. The following facts are known about their game:

1. At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
2. At the end of six rounds, Arun had scored 7 points, Bankim and Dipak had scored -1 point each, and Charu had scored -5 points.
3. Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
4. In exactly two out of the six rounds, Arun was the only player who bid Hi.

## Question 41

In how many rounds did Dipak gain exactly 1 point?

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak played the Hi-Lo game. The following facts are known about their game:

1. At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
2. At the end of six rounds, Arun had scored 7 points, Bankim and Dipak had scored -1 point each, and Charu had scored -5 points.
3. Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
4. In exactly two out of the six rounds, Arun was the only player who bid Hi.

Case 1: All Hi → -1 each

Case 2: 3 Hi, 1 Lo → +1 each for Hi, -3 for Lo

Case 3: 2 Hi, 2 Lo → +2 each for Hi, -2 each for Lo

Case 4: 1 Hi, 3 Lo → +3 for Hi, -1 each for Lo

Case 5: All Lo → +1 each

## Question 42

In how many rounds did Arun bid Hi?

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak played the Hi-Lo game. The following facts are known about their game:

1. At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
2. At the end of six rounds, Arun had scored 7 points, Bankim and Dipak had scored -1 point each, and Charu had scored -5 points.
3. Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
4. In exactly two out of the six rounds, Arun was the only player who bid Hi.

Case 1: All Hi → -1 each

Case 2: 3 Hi, 1 Lo → +1 each for Hi, -3 for Lo

Case 3: 2 Hi, 2 Lo → +2 each for Hi, -2 each for Lo

Case 4: 1 Hi, 3 Lo → +3 for Hi, -1 each for Lo

Case 5: All Lo → +1 each



## Question 43

In how many rounds did Bankim bid Lo?

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak played the Hi-Lo game. The following facts are known about their game:

- At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
- At the end of six rounds, Arun had scored 7 points, Bankim had scored -1 point each, and Charu had scored -5 points.
- Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
- In exactly two out of the six rounds, Arun was the only player who bid Hi.

*Case 1:*

|    |   |    |
|----|---|----|
| Hi | } | -1 |
| Hi |   |    |
| Hi |   |    |
| Lo |   |    |

*Case 2:*

|    |    |    |
|----|----|----|
| Hi | }  | +1 |
| Hi |    |    |
| Hi |    |    |
| Lo | -3 |    |

*Case 3:*

|    |    |    |
|----|----|----|
| Hi | }  | +2 |
| Hi |    |    |
| Lo |    |    |
| Lo | -2 |    |

*Case 4:*

|    |    |    |
|----|----|----|
| Hi | }  | +3 |
| Lo |    |    |
| Lo |    |    |
| Lo | -1 |    |

VIDEO SOLUTION

## Question 44

In how many rounds did all four players make identical bids?

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak played the Hi-Lo game. The following facts are known about their game:

- At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
- At the end of six rounds, Arun had scored 7 points, Bankim had scored -1 point each, and Charu had scored -5 points.
- Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
- In exactly two out of the six rounds, Arun was the only player who bid Hi.

*Case 1:*

|    |   |    |
|----|---|----|
| Hi | } | -1 |
| Hi |   |    |
| Hi |   |    |
| Lo |   |    |

*Case 2:*

|    |    |    |
|----|----|----|
| Hi | }  | +1 |
| Hi |    |    |
| Hi |    |    |
| Lo | -3 |    |

*Case 3:*

|    |    |    |
|----|----|----|
| Hi | }  | +2 |
| Hi |    |    |
| Lo |    |    |
| Lo | -2 |    |

*Case 4:*

|    |    |    |
|----|----|----|
| Hi | }  | +3 |
| Lo |    |    |
| Lo |    |    |
| Lo | -1 |    |

VIDEO SOLUTION

## Question 45

What were the bids by Arun, Bankim, Charu and Dipak, respectively in the first round?

- A Hi, Lo, Lo, Hi
- B Hi, Lo, Lo, Lo
- C Hi, Hi, Lo, Lo
- D Lo, Lo, Lo, Hi

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak play the Hi-Lo game. The following facts are known about their game:

1. At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
2. At the end of six rounds, Arun had scored 7 points, Bankim had scored -1 point each, and Charu had scored -5 points.
3. Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
4. In exactly two out of the six rounds, Arun was the only player who bid Hi.

Case 1) Hi: -1, Hi: -1, Hi: -1, Lo: -3  
Case 2) Hi: +1, Hi: +1, Hi: +1, Lo: -3  
Case 3) Hi: +2, Hi: +2, Lo: -2, Lo: -2  
Case 4) Hi: +3, Lo: -1, Lo: -1, Lo: -1



VIDEO SOLUTION

## 100+ CAT Quant Questions



### Question 46

In which of the following rounds, was Arun DEFINITELY the only player to bid Hi?

- A Second
- B Third
- C Fourth
- D First

**cracku**

The Hi-Lo game is a four-player game played in six rounds. In every round, each player chooses to bid Hi or Lo. The bids are made simultaneously. If all four bid Hi, then all four lose 1 point each. If three players bid Hi and one bids Lo, then the players bidding Hi gain 1 point each and the player bidding Lo loses 3 points. If two players bid Hi and two bid Lo, then the players bidding Hi gain 2 points each and the players bidding Lo lose 2 points each. If one player bids Hi and three bid Lo, then the player bidding Hi gains 3 points and the players bidding Lo lose 1 point each. If all four bid Lo, then all four gain 1 point each. Four players Arun, Bankim, Charu, and Dipak play the Hi-Lo game. The following facts are known about their game:

1. At the end of three rounds, Arun had scored 6 points, Dipak had scored 2 points, Bankim and Charu had scored -2 points each.
2. At the end of six rounds, Arun had scored 7 points, Bankim had scored -1 point each, and Charu had scored -5 points.
3. Dipak's score in the third round was less than his score in the first round but was more than his score in the second round.
4. In exactly two out of the six rounds, Arun was the only player who bid Hi.

Case 1) Hi: -1, Hi: -1, Hi: -1, Lo: -3  
Case 2) Hi: +1, Hi: +1, Hi: +1, Lo: -3  
Case 3) Hi: +2, Hi: +2, Lo: -2, Lo: -2  
Case 4) Hi: +3, Lo: -1, Lo: -1, Lo: -1



VIDEO SOLUTION

### Instructions

The game of Chango is a game where two people play against each other; one of them wins and the other loses, i.e., there are no drawn Chango games. 12 players participated in a Chango championship. They were divided into four groups: Group A consisted of Aruna, Azul, and Arif; Group B consisted of Brinda, Brij, and Biju; Group C consisted of Chitra, Chetan, and Chhavi; and Group D consisted of Dipen, Donna, and Deb.

Players within each group had a distinct rank going into the championship. The players have NOT been listed necessarily according to their ranks. In the group stage of the game, the second and third ranked players play against each other, and the winner of that game plays against the first ranked player of the group. The winner of this second game is considered as the winner of the group and enters a semi-final.

The winners from Groups A and B play against each other in one semi-final, while the winners from Groups C and D play against each other in the other semi-final. The winners of the two semi-finals play against each other in the final to decide the winner of the championship.

It is known that:

1. Chitra did not win the championship.
2. Aruna did not play against Arif. Brij did not play against Brinda.
3. Aruna, Biju, Chitra, and Dipen played three games each, Azul and Chetan played two games each, and the remaining players played one game each.

#### Question 47

Who won the championship?

- A Chitra
- B Aruna
- C Brij
- D Cannot be determined

VIDEO SOLUTION

#### Question 48

Which of the following pairs must have played against each other in the championship?

- A Deb, Donna
- B Azul, Biju
- C Donna, Chetan
- D Chitra, Dipen



cracku



Who among the following did NOT play against Chitra in the championship?

-  VIDEO SOLUTION

Who among the following was DEFINITELY NOT ranked first in his/her group?

- 28/115

cracku

Players within each group had a distinct rank going into the championship. The players have NOT been listed necessarily according to their ranks. In the group stage of the game, the second and third ranked players play against each other, and the winner of that game plays against the first ranked player of the group. The winner of this second game is considered as the winner of the group and enters a semi-final. The winners from Groups A and B play against each other in one semi-final, while the winners from Groups C and D play against each other in the other semi-final. The winners of the two semi-finals play against each other in the final to decide the champion.

It is known that:  
 1. Chitra did not win the championship.  
 Arif did not play against Brinda.  
 Chitra did not play against Arif.  
 Chitra and Arif played three games each.  
 Arif and Chetan played one game each.

VIDEO SOLUTION

### Instructions

For the following questions answer them individually

#### Question 51

A, B, C, D, ..., X, Y, Z are the players who participated in a tournament. Everyone played with every other player exactly once. A win scores 2 points, a draw scores 1 point and a loss scores 0 point. None of the matches ended in a draw. No two players scored the same score. At the end of the tournament, by ranking list is published which is in accordance with the alphabetical order. Then

- A M wins over N
- B N wins over M
- C M does not play with N
- D None of these

VIEW SOLUTION

## 100+ CAT VARC Questions



#### Question 52

116 people participated in a singles tennis tournament of knock out format. The players are paired up in the first round, the winners of the first round are paired up in second round, and so on till the final is played between two players. If after any round, there is odd number of players, one player is given a bye, i.e. he skips that round and plays the next round with the winners. Find the total number of matches played in the tournament.

- A 115
- B 53
- C 232
- D 116

VIEW SOLUTION

### Instructions

Pulak, Qasim, Ritesh, and Suresh participated in a tournament comprising of eight rounds. In each round, they formed two pairs, with each of them being in exactly one pair. The only restriction in the pairing was that the pairs would change in successive rounds. For example, if Pulak formed a pair with Qasim in the first round, then he would have to form a pair with Ritesh or Suresh in the second round. He would be free to pair with Qasim again in the third round. In each round, each pair decided whether to play the game in that round or not. If they decided not to play, then no money was exchanged between them. If they decided to play, they had to bet either ₹1 or ₹2 in that round. For example, if they chose to bet ₹2, then the player winning the game got ₹2 from the one losing the game.

At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 8 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 |       | ₹8    | ₹10    | ₹10    |
| Round 2 | ₹13   | ₹10   |        | ₹8     |
| Round 3 |       |       |        | ₹10    |
| Round 4 |       |       |        |        |
| Round 5 | ₹10   | ₹10   |        | ₹13    |
| Round 6 |       |       |        |        |
| Round 7 |       | ₹12   | ₹4     |        |
| Round 8 | ₹13   |       |        | ₹10    |

### Question 53

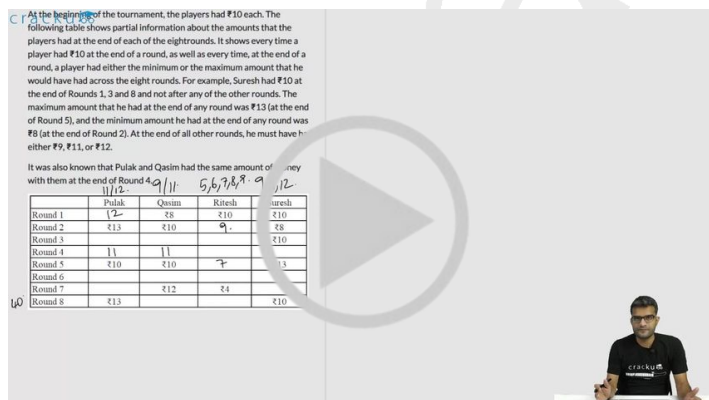
What BEST can be said about the amount of money that Pulak had with him at the end of Round 6?

- A Exactly ₹12
- B Exactly ₹11
- C ₹12 or ₹13
- D ₹11 or ₹12

At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 8 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 |       | ₹8    | ₹10    | ₹10    |
| Round 2 | ₹13   | ₹10   |        | ₹8     |
| Round 3 |       |       |        | ₹10    |
| Round 4 |       |       |        |        |
| Round 5 | ₹10   | ₹10   |        | ₹13    |
| Round 6 |       |       |        |        |
| Round 7 |       | ₹12   | ₹4     |        |
| Round 8 | ₹13   |       |        | ₹10    |



VIDEO SOLUTION

### Question 54

How much money (in ₹) did Ritesh have at the end of Round 4?

At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 8 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        | 10     |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 12    | 14     |        |
| Round 7 |       |       |        |        |
| Round 8 | 13    |       |        | 10     |

VIDEO SOLUTION

## CAT Expected Questions PDF



### Question 55

How many games were played with a bet of ₹2?

At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 8 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        | 10     |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 12    | 14     |        |
| Round 7 |       |       |        |        |
| Round 8 | 13    |       |        | 10     |

VIDEO SOLUTION

### Question 56

Which of the following pairings was made in Round 5?

- A Qasim and Suresh
- B Pulak and Ritesh
- C Pulak and Qasim
- D Pulak and Suresh



At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 6 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        | 10     |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

VIDEO SOLUTION

### Question 57

What BEST can be said about the amount of money that Ritesh had with him at the end of Round 8?

- A ₹4 or ₹5
- B Exactly ₹5
- C ₹5 or ₹6
- D Exactly ₹6

At the beginning of the tournament, the players had ₹10 each. The following table shows partial information about the amounts that the players had at the end of each of the eight rounds. It shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds. For example, Suresh had ₹10 at the end of Rounds 1, 3 and 6 and not after any of the other rounds. The maximum amount that he had at the end of any round was ₹13 (at the end of Round 5), and the minimum amount he had at the end of any round was ₹8 (at the end of Round 2). At the end of all other rounds, he must have had either ₹9, ₹11, or ₹12.

It was also known that Pulak and Qasim had the same amount of money with them at the end of Round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        | 10     |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

VIDEO SOLUTION

## CAT Important Topics

### Instructions

Ten players, as listed in the table below, participated in a rifle shooting competition comprising of 10 rounds. Each round had 6 participants. Players numbered 1 through 6 participated in Round 1, players 2 through 7 in Round 2,..., players 5 through 10 in Round 5, players 6 through 10 and 1 in Round 6, players 7 through 10, 1 and 2 in Round 7 and so on. The top three performances in each round were awarded 7, 3 and 1 points respectively. There were no ties in any of the 10 rounds. The table below gives the total number of points obtained by the 10 players after Round 6 and Round 10.

| Player No. | Player Name | Points after Round 6 | Points after Round 10 |
|------------|-------------|----------------------|-----------------------|
| 1          | Amita       | 8                    | 18                    |
| 2          | Bala        | 2                    | 5                     |
| 3          | Chen        | 3                    | 6                     |
| 4          | David       | 6                    | 6                     |
| 5          | Eric        | 3                    | 10                    |
| 6          | Fatima      | 10                   | 10                    |
| 7          | Gordon      | 17                   | 17                    |
| 8          | Hansa       | 1                    | 4                     |
| 9          | Ikea        | 2                    | 17                    |
| 10         | Joshin      | 14                   | 17                    |

The following information is known about Rounds 1 through 6:

1. Gordon did not score consecutively in any two rounds.
2. Eric and Fatima both scored in a round.

The following information is known about Rounds 7 through 10:

1. Only two players scored in three consecutive rounds. One of them was Chen. No other player scored in any two consecutive rounds.
2. Joshin scored in Round 7, while Amita scored in Round 10.
3. No player scored in all the four rounds.

#### Question 58

Which players scored points in the last round?

- A Amita, Eric, Joshin
- B Amita, Chen, David
- C Amita, Bala, Chen
- D Amita, Chen, Eric

cracku

Ten players, as listed in the table below, participated in a rifle shooting competition comprising of 10 rounds. Each round had 6 participants. Players numbered 1 through 6 participated in Round 1, players 2 through 7 in Round 2, ... players 5 through 10 in Round 5, players 6 through 10 and 1 in Round 6, players 7 through 10, 1 and 2 in Round 7 and so on. The top three performances in each round were awarded 7, 3 and 1 points respectively. There were no ties in any of the 10 rounds. The table below gives the total number of points obtained by the 10 players after Round 6 and Round 10.

| Player No. | Player Name | Points after Round 6 | Points after Round 10 |
|------------|-------------|----------------------|-----------------------|
| 1          | Amita       | 8                    | 18                    |
| 2          | Bala        | 2                    | 5                     |
| 3          | Chen        | 3                    | 6                     |
| 4          | David       | 6                    | 6                     |
| 5          | Eric        | 3                    | 10                    |
| 6          | Fatima      | 10                   | 10                    |
| 7          | Gordon      | 17                   | 17                    |
| 8          | Hansa       | 1                    | 4                     |
| 9          | Ikea        | 2                    | 17                    |
| 10         | Joshin      | 14                   | 17                    |

The following information is known about Rounds 1 through 6:

1. Gordon did not score consecutively in any two rounds.
2. Eric and Fatima both scored in a round.

The following information is known about Rounds 7 through 10:

1. Only two players scored in three consecutive rounds. One of them was Chen. No other player scored in any two consecutive rounds.
2. Joshin scored in Round 7, while Amita scored in Round 10.

VIDEO SOLUTION

#### Question 59

Which three players were in the last three positions after Round 4?

- A Bala, Ikea, Joshin
- B Bala, Hansa, Ikea
- C Bala, Chen, Gordon
- D Hansa, Ikea, Joshin

cracku

Ten players, as listed in the table below, participated in a rifle shooting competition comprising of 10 rounds. Each round had 6 participants. Players numbered 1 through 6 participated in Round 1, players 2 through 7 in Round 2, ..., players 5 through 10 in Round 5, players 6 through 10 and 1 in Round 6, players 7 through 10, 1 and 2 in Round 7 and so on. The top three performances in each round were awarded 7, 3 and 1 points respectively. There were no ties in any of the 10 rounds. The table below gives the total number of points obtained by the 10 players after Round 6 and Round 10.

| Player No. | Player Name | Points after Round 6 | Points after Round 10 |
|------------|-------------|----------------------|-----------------------|
| 1          | Amita       | 8                    | 18                    |
| 2          | Bala        | 2                    | 5                     |
| 3          | Chen        | 3                    | 6                     |
| 4          | David       | 6                    | 1                     |
| 5          | Eric        | 3                    | 1                     |
| 6          | Fatima      | 10                   | 1                     |
| 7          | Gordon      | 17                   | 17                    |
| 8          | Hansa       | 1                    | 4                     |
| 9          | Ikea        | 2                    | 17                    |
| 10         | Joshin      | 14                   | 17                    |

The following information is known about Rounds 1 through 6:

- Gordon did not score consecutively in any two rounds.
- Eric and Fatima both scored in a round.

The following information is known about Rounds 7 through 10:

- Only two players scored in three consecutive rounds. One of them was Chen. No other player scored in any two consecutive rounds.
- Joshin scored in Round 7, while Amita scored in Round 10.

VIDEO SOLUTION

### Question 60

What were the scores of Chen, David, and Eric respectively after Round 3?

- A 3, 6, 3
- B 3, 3, 3
- C 3, 3, 0
- D 3, 0, 3

cracku

Ten players, as listed in the table below, participated in a rifle shooting competition comprising of 10 rounds. Each round had 6 participants. Players numbered 1 through 6 participated in Round 1, players 2 through 7 in Round 2, ..., players 5 through 10 in Round 5, players 6 through 10 and 1 in Round 6, players 7 through 10, 1 and 2 in Round 7 and so on. The top three performances in each round were awarded 7, 3 and 1 points respectively. There were no ties in any of the 10 rounds. The table below gives the total number of points obtained by the 10 players after Round 6 and Round 10.

| Player No. | Player Name | Points after Round 6 | Points after Round 10 |
|------------|-------------|----------------------|-----------------------|
| 1          | Amita       | 8                    | 18                    |
| 2          | Bala        | 2                    | 5                     |
| 3          | Chen        | 3                    | 6                     |
| 4          | David       | 6                    | 1                     |
| 5          | Eric        | 3                    | 1                     |
| 6          | Fatima      | 10                   | 1                     |
| 7          | Gordon      | 17                   | 17                    |
| 8          | Hansa       | 1                    | 4                     |
| 9          | Ikea        | 2                    | 17                    |
| 10         | Joshin      | 14                   | 17                    |

The following information is known about Rounds 1 through 6:

- Gordon did not score consecutively in any two rounds.
- Eric and Fatima both scored in a round.

The following information is known about Rounds 7 through 10:

- Only two players scored in three consecutive rounds. One of them was Chen. No other player scored in any two consecutive rounds.
- Joshin scored in Round 7, while Amita scored in Round 10.

VIDEO SOLUTION

cracku

**CAT WhatsApp Group**

WhatsApp icon

### Question 61

Which player scored points in maximum number of rounds?

- A Joshin
- B Chen
- C Amita
- D Ikea

Ten players, as listed in the table below, participated in a rifle shooting competition comprising of 10 rounds. Each round had 6 participants. Players numbered 1 through 6 participated in Round 1, players 2 through 7 in Round 2, ..., players 5 through 10 in Round 5, players 6 through 10 and 1 in Round 6, players 7 through 10, 1 and 2 in Round 7 and so on. The top three performances in each round were awarded 7, 3 and 1 points respectively. There were no ties in any of the 10 rounds. The table below gives the total number of points obtained by the 10 players after Round 6 and Round 10.

| Player No. | Player Name | Points after Round 6 | Points after Round 10 |
|------------|-------------|----------------------|-----------------------|
| 1          | Amita       | 8                    | 18                    |
| 2          | Bala        | 2                    | 5                     |
| 3          | Chen        | 3                    | 6                     |
| 4          | David       | 6                    | 1                     |
| 5          | Eric        | 3                    | 1                     |
| 6          | Fatima      | 10                   | 11                    |
| 7          | Gordon      | 17                   | 17                    |
| 8          | Hansa       | 1                    | 4                     |
| 9          | Ikes        | 2                    | 17                    |
| 10         | Joshin      | 14                   | 17                    |

The following information is known about Rounds 1 through 6:

1. Gordon did not score consecutively in any two rounds.

2. Eric and Fatima both scored in a round.

The following information is known about Rounds 7 through 10:

1. Only two players scored in three consecutive rounds. One of them was Chen. No other player scored in any two consecutive rounds.

2. Joshin scored in Round 7, while Amita scored in Round 10.

1 2 3 4

A  
B  
C  
D  
E  
F  
G  
H  
I  
J



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## Answers

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 1.5  | 2.A  | 3.8  | 4.3  | 5.4  | 6.B  | 7.A  | 8.C  |
| 9.6  | 10.D | 11.A | 12.7 | 13.E | 14.B | 15.D | 16.E |
| 17.D | 18.B | 19.D | 20.D | 21.D | 22.A | 23.C | 24.C |
| 25.C | 26.D | 27.C | 28.C | 29.A | 30.C | 31.D | 32.C |
| 33.B | 34.C | 35.C | 36.C | 37.D | 38.C | 39.A | 40.B |
| 41.1 | 42.4 | 43.4 | 44.2 | 45.A | 46.A | 47.B | 48.D |
| 49.D | 50.A | 51.A | 52.A | 53.A | 54.6 | 55.6 | 56.D |
| 57.D | 58.D | 59.D | 60.B | 61.D |      |      |      |

## Explanations

1.5

We are told that there are six teams, that are divided into two groups.

Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be  ${}^3C_2$  games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this, From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is  $3 \times 3$  which is 9. And since they play each other twice, that is  $9 \times 2$  games of this combination.

Total number of games is  $6 + 18 = 24$

It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to be played in this format, the number of rounds will be  $24/3 = 8$

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

| Group-1 | Group-2 |
|---------|---------|
| 1       | 2       |
| 5       | 3       |
| 6       | 4       |

Now, using the given information to build the matchups for the 8 rounds.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         |         |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
|         | 1 vs 6  |         | 3 vs 6  |
|         |         |         | 2 vs 5  |
|         |         |         |         |

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         | 2 vs 3  |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  |         | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,

We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
|         | 1 vs 5  | 5 vs 6  |         |
|         | 2 vs 3  | 1 vs 2  |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.







| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
| 1 vs 2  | 1 vs 5  | 5 vs 6  | 1 vs 3  |
| 3 vs 5  | 2 vs 3  | 1 vs 2  | 4 vs 5  |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  | 2 vs 4  | 1 vs 3  | 2 vs 5  |
| 1 vs 4  | 3 vs 5  | 4 vs 5  | 1 vs 4  |

. This is the final table.

2, 3 and 4 were part of the same group. 5 is the answer.

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### 3.8

We are told that there are six teams, that are divided into two groups.

Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be  ${}^3C_2$  games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,

From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is  $3 \times 3$  which is 9. And since they play each other twice, that is  $9 \times 2$  games of this combination.

Total number of games is  $18 + 6 = 24$

It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to be played in this format, the number of rounds will be  $24/3 = 8$

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

| Group-1 | Group-2 |
|---------|---------|
| 1       | 2       |
| 5       | 3       |
| 6       | 4       |

Now, using the given information to build the matchups for the 8 rounds.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         |         |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
|         | 1 vs 6  |         | 3 vs 6  |
|         |         |         | 2 vs 5  |
|         |         |         |         |

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         | 2 vs 3  |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  |         | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,

We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
|         | 1 vs 5  | 5 vs 6  |         |
|         | 2 vs 3  | 1 vs 2  |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
| 1 vs 2  | 1 vs 5  | 5 vs 6  | 1 vs 3  |
| 3 vs 5  | 2 vs 3  | 1 vs 2  | 4 vs 5  |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  | 2 vs 4  | 1 vs 3  | 2 vs 5  |
| 1 vs 4  | 3 vs 5  | 4 vs 5  | 1 vs 4  |

. This is the final table.

The tournament will have 8 rounds.

[VIDEO SOLUTION](#)

#### 4.3

We are told that there are six teams, that are divided into two groups.

Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be  ${}^3C_2$  games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,

From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is  $3 \times 3$  which is 9. And since they play each other twice, that is  $9 \times 2$  games of this combination.

Total number of games is  $18 + 6 = 24$

It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to be played in this format, the number of rounds will be  $24/3 = 8$

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

| Group-1 | Group-2 |
|---------|---------|
| 1       | 2       |
| 5       | 3       |
| 6       | 4       |

Now, using the given information to build the matchups for the 8 rounds.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         |         |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
|         | 1 vs 6  |         | 3 vs 6  |
|         |         |         | 2 vs 5  |
|         |         |         |         |

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         | 2 vs 3  |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  |         | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,

We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
|         | 1 vs 5  | 5 vs 6  |         |
|         | 2 vs 3  | 1 vs 2  |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
| 1 vs 2  | 1 vs 5  | 5 vs 6  | 1 vs 3  |
| 3 vs 5  | 2 vs 3  | 1 vs 2  | 4 vs 5  |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  | 2 vs 4  | 1 vs 3  | 2 vs 5  |
| 1 vs 4  | 3 vs 5  | 4 vs 5  | 1 vs 4  |

. This is the final table.

The team that played Team 1 in Round 7 is 3.

[VIDEO SOLUTION](#)

#### 5.4

We are told that there are six teams, that are divided into two groups.

Teams in the same group will play each other only once, and teams in different group will play each other twice.

Calculating the combinations, there is going to be  ${}^3C_2$  games among teams in the same group among them, and since there is two groups, total such games will be 6.

Now, teams in different group play each other twice. Calculating the combinations for this,

From the first group, a team can be chosen in three ways, and from the second group, a team can be chosen in three ways. Total ways two teams from different groups can play each other is  $3 \times 3$  which is 9. And since they play each other twice, that is  $9 \times 2$  games of this combination.

Total number of games is  $18 + 6 = 24$

It is given that each team plays one game in each round, that means there is going to be 3 matchups in each round. And given, there is 24 games to be played in this format, the number of rounds will be  $24/3 = 8$

The tournament will have 8 rounds.

Now that we know there are going to be 8 rounds in the tournament, let us identify the teams in a particular group, that will help us build the matchups.

We are told that Round 8 teams from different groups play each other, and Teams 1 and 5 play only once. This means, 1 and 5 have to be on the same group. It is also told that 4 and 6 play each other twice, that means 4 and 6 have to be in different groups. Looking at the matches from Round 8 that is given to us, 3 played 6 and 2 played 5. We already know 1 and 5 are in the same group, so 2 must be in the other group. Among 3 and 6, if we were to place 3 in the group with 1 and 5, 4 and 6 would have to be in the same group, which is not possible, hence 6 is with 1 and 5, giving us the final combination of groups.

| Group-1 | Group-2 |
|---------|---------|
| 1       | 2       |
| 5       | 3       |
| 6       | 4       |

Now, using the given information to build the matchups for the 8 rounds.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         |         |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
|         | 1 vs 6  |         | 3 vs 6  |
|         |         |         | 2 vs 5  |
|         |         |         |         |

Rounds marked with the same colour represent the fact that the matchups are identical. Now we know that each team plays a game in each round, we know 2 out of the 3 matches for Round 2 and 8, and we can identify the third matchups as well. Giving us this resulting table.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  |         |
|         | 1 vs 5  |         |         |
|         | 2 vs 3  |         |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  |         | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

We are told that Round 4 and 7 are identical, that means they are the matchups between teams from two different groups,

We look at the matchups that are remaining among the 6 teams where both the games are left to play.

Right away we identify that 6 is yet to play 2 twice and 5 once. We are looking for teams playing twice, so both Round 4 and 7 has a matchups between 2 and 6. This means 6 will play 5 in Round 3 and using that we can identify the third matchup in Round 3 as well.

| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
|         | 1 vs 5  | 5 vs 6  |         |
|         | 2 vs 3  | 1 vs 2  |         |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  |         |         | 2 vs 5  |
| 1 vs 4  |         |         | 1 vs 4  |

Now, we can identify that 1 is yet to play 3 both the times, 2 once. And we are looking for teams playing each other twice, so we can fill in the remaining values in the table as well.



| Round-1 | Round-2 | Round-3 | Round-4 |
|---------|---------|---------|---------|
| 4 vs 6  | 4 vs 6  | 3 vs 4  | 2 vs 6  |
| 1 vs 2  | 1 vs 5  | 5 vs 6  | 1 vs 3  |
| 3 vs 5  | 2 vs 3  | 1 vs 2  | 4 vs 5  |
| Round-5 | Round-6 | Round-7 | Round-8 |
| 3 vs 6  | 1 vs 6  | 2 vs 6  | 3 vs 6  |
| 2 vs 5  | 2 vs 4  | 1 vs 3  | 2 vs 5  |
| 1 vs 4  | 3 vs 5  | 4 vs 5  | 1 vs 4  |

. This is the final table.

The team that played Team 1 in Round 5 is 4.

[VIDEO SOLUTION](#)



6.B

After second move i will be having from a quarter cell of board to make a move.

Since left quarters have cells with values 1, Hence i will choose to retain right in my first move so that second person can force my profit to be either 2 or 3 not 1.

So now after my first move of retaining right, we have two quarters

Now intelligently he will make a move to retain with lower half so that at fourth move i can retain with choices of 2 and 3 not 2 and 4.

[VIDEO SOLUTION](#)

7.A

After retaining right in first move, second person will have 2 quarters to choose in which minimum value is 2 and 3.

If he retains upper half, i will retain right as to maximize my profit of either 4 or 7.

If he retains lower half, i will retain left as to maximize my profit of either 3 or 8.

Hence whatever he retains, i won't have a profit less than 3.

[VIDEO SOLUTION](#)

8.C

After my first move of retaining right board will be like

|   |   |
|---|---|
| 2 | 4 |
| 6 | 7 |
| 3 | 2 |
| 8 | 4 |

Now second person will force me to retain with the lowest value possible i.e. 2 or 3

So if he retains lower half and i retain with left one (as mentioned) then he can only force me to pick 3.

Or if he retains upper half and i retain with left one(as mentioned) then he can retain upper half so i will be

having minimized profit of 2.

Hence second person will have moves as "Retain (upper half) and retain (upper half)"

 VIDEO SOLUTION

9.6

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

$$6 + \text{Badal} + \text{Clive} + 11 + 9 = 40$$

$$\text{Badal} + \text{Clive} = 14$$

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be  $1 + 1 + 2 + 2 = 6$ . It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       |       |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       |       |       |       |       | 9    |
| TAPS RECEIVED | 9     | a     |       | a     | a     | 40   |

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as  $2 + 2 + 1 + 3 = 8$ . We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       | 1/3   |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       | 3/1   |       |       |       | 9    |
| TAPS RECEIVED | 9     | 8     | 7     | 8     | 8     | 40   |

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be  $9 - 3 - 1 = 5$ . So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is  $8 - 1 = 7$ . So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

|                      | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|----------------------|-------|-------|-------|-------|-------|------|
| <b>A</b>             |       | 2     | 1     | 1     | 2     | 6    |
| <b>B</b>             | 1     |       | 2/1   | 2     | 1/2   | 6    |
| <b>C</b>             | 2     | 1/3   |       | 2/3   | 2/1   | 8    |
| <b>D</b>             | 3     | 2     | 3     |       | 3     | 11   |
| <b>E</b>             | 3     | 3/1   | 1/2   | 3/2   |       | 9    |
| <b>TAPS RECEIVED</b> | 9     | 8     | 7     | 8     | 8     | 40   |

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,

|                      | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|----------------------|-------|-------|-------|-------|-------|------|
| <b>A</b>             |       | 2     | 1     | 1     | 2     | 6    |
| <b>B</b>             | 1     |       | 2     | 2     | 1     | 6    |
| <b>C</b>             | 2     | 1     |       | 3     | 2     | 8    |
| <b>D</b>             | 3     | 2     | 3     |       | 3     | 11   |
| <b>E</b>             | 3     | 3     | 1     | 2     |       | 9    |
| <b>TAPS RECEIVED</b> | 9     | 8     | 7     | 8     | 8     | 40   |

The total number of Yes responses is equal to the number of 1's in the table, which is 6.

Hence, the correct answer is 6.

[VIEW SOLUTION](#)

#### 10. D

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

$$6 + \text{Badal} + \text{Clive} + 11 + 9 = 40$$

$$\text{Badal} + \text{Clive} = 14$$

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be  $1 + 1 + 2 + 2 = 6$ . It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The

sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       |       |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       |       |       |       |       | 9    |
| TAPS RECEIVED | 9     | a     |       | a     | a     | 40   |

Alia and Badal are the only people with an equal number of taps.

Hence, the correct answer is option D.

[VIEW SOLUTION](#)



11. A

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

$$6 + \text{Badal} + \text{Clive} + 11 + 9 = 40$$

$$\text{Badal} + \text{Clive} = 14$$

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be  $1 + 1 + 2 + 2 = 6$ . It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       |       |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       |       |       |       |       | 9    |
| TAPS RECEIVED | 9     | a     |       | a     | a     | 40   |

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as  $2 + 2 + 1 + 3 = 8$ . We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       | 1/3   |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       | 3/1   |       |       |       | 9    |
| TAPS RECEIVED | 9     | 8     | 7     | 8     | 8     | 40   |

We are given that the answer by Alia does not match the answers people gave to her question. We know that the taps by Badal have to be 1, 1, 2 and 2 in some order. We know that Alia tapped twice to Badal's question, so Badal cannot tap twice to Alia's question, which means Badal tapped once for Alia's question. Now the sum of Clive and Ehsan's taps for Alia's question has to be  $9 - 3 - 1 = 5$ . So, they have to be 2 and 3 in some order. We also know that Alia's answer to Ehsan's question is 2, so Ehsan has only one option to answer Alia, which is 3, and Clive's answer to Alia's question is 2.

Badal and Ehsan's answer to Clive's question has to be 1 and 2 in some order. Similarly, Badal and Clive's answers to Ehsan's question have to be 1 and 3 in some order.

We can also calculate the sum of taps by Badal, Clive and Ehsan to Dilshan's question is  $8 - 1 = 7$ . So, their taps have to be 2, 2 and 3 in some order as this is the only possibility satisfying the condition of at least one Yes, one No and one Maybe for every question. We know that a tap Badal has to be either one or two, so the only possibility for Badal's answer to Dilshan's question is 2 and the other two has to be two and three in some order.

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             | 1     |       | 2/1   | 2     | 1/2   | 6    |
| C             | 2     | 1/3   |       | 2/3   | 2/1   | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             | 3     | 3/1   | 1/2   | 3/2   |       | 9    |
| TAPS RECEIVED | 9     | 8     | 7     | 8     | 8     | 40   |

We are also given that the answer by Ehsan does not match the answers people gave to his question. Dilshan's answer to Ehsan's question is 3, so the answer by Ehsan to Dilshan's question has to be 2.

The answer by Clive to Badal's question has to be 1, as if it is 3, then the answer to Ehsan's question by Clive becomes 0, which is not possible. With that value, we can determine all the other values and putting the values in the table, we get,



|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             | 1     |       | 2     | 2     | 1     | 6    |
| C             | 2     | 1     |       | 3     | 2     | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             | 3     | 3     | 1     | 2     |       | 9    |
| TAPS RECEIVED | 9     | 8     | 7     | 8     | 8     | 40   |

Clive's answer to Ehsan's question is No, which is denoted by 2.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

### 12.7

We are given that Alia tapped 6 times, Dilshan tapped 11 times, and Ehsan tapped 9 times. We also know that the total number of taps is 40. So, the sum of taps by Badal and Clive can be calculated as,

$$6 + \text{Badal} + \text{Clive} + 11 + 9 = 40$$

$$\text{Badal} + \text{Clive} = 14$$

We are given that taps by Clive are more than taps by Badal.

So, the possible pairs for taps by Clive and Badal such that the sum is 14 are (8, 6), (9, 5), (10, 4)...(14, 0).

We are given that no one gave more than two 'Yes' answers. We know that yes means one tap. So, the maximum number of 1 taps out of the 4 questions answered can be 2. The minimum possible number of taps for any person for the four questions would be  $1 + 1 + 2 + 2 = 6$ . It is not possible for any person to have fewer than 6 taps as the maximum number of yes is 2.

So, out of all the possibilities, the only possible case for Badal is 6, and Clive is 8, as in all the other cases, the number by Badal was less than 6.

We are given that Alia answered Yes to Clive and Dilshan's questions. We know that the total number of taps by Alia is 6, and she tapped once for both those questions. The number of taps by Alia to the questions by Badal and Ehsan must be greater than 1, as we know that a person must have a maximum of two single taps. The sum of taps by Alia to Badal and Ehsan can be calculated as,

$$\text{Badal} + 1 + 1 + \text{Ehsan} = 6$$

$$\text{Badal} + \text{Ehsan} = 4$$

We know that both the values are greater than 1 and their sum is 4, so the only possibility is for both of them to be equal to 2.

So, Alia tapped twice for both Badal and Ehsan.

We are given that Dilshan answered no to Badal's question which means he tapped twice. The total number of taps by Dilshan is 11. So, the sum of taps from Alia, Clive and Ehsan's questions can be calculated as,

$$\text{Alia} + 2 + \text{Clive} + \text{Ehsan} = 11$$

$$\text{Alia} + \text{Clive} + \text{Ehsan} = 9$$

We know that the maximum possibility for each of the values is 3, and for the sum to be 9, all the values must be equal to 3.

We are given that taps received for the questions by Badal, Dilshan and Ehsan are equal, so let us assume the value to be a.

Putting all the calculated values in the table, we get,

|               | Q1(A) | Q2(B) | Q3(C) | Q4(D) | Q5(E) | TAPS |
|---------------|-------|-------|-------|-------|-------|------|
| A             |       | 2     | 1     | 1     | 2     | 6    |
| B             |       |       |       |       |       | 6    |
| C             |       |       |       |       |       | 8    |
| D             | 3     | 2     | 3     |       | 3     | 11   |
| E             |       |       |       |       |       | 9    |
| TAPS RECEIVED | 9     | a     |       | a     | a     | 40   |

We know that each question received atleast one yes, one no and one maybe as an answer. If we examine the question asked by B, we already have two 'No's, so out of the other two, one must be 'Yes', and the other must be 'Maybe'. The taps received by B can be calculated as  $2 + 2 + 1 + 3 = 8$ . We calculated the value of a to be 8, and the value of taps received by C can be calculated as,

$$9 + 8 + c + 8 + 8 = 40$$

$$c = 7$$

So, the number of taps received by Clive is 7.

Hence, the correct answer is 7.

[VIEW SOLUTION](#)

### 13. E

There are a total of  ${}^6C_2$  matches  $\Rightarrow$  15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the wins by each team are A(3), B(4), C(0), D(2), E(4), F(2). Hence, most wins are by B and E.

[VIDEO SOLUTION](#)

### 14. B

There are a total of  ${}^6C_2$  matches  $\Rightarrow$  15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D, E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the two teams that won against stage 1 leader A are E and F.

 VIDEO SOLUTION

15. D

There are a total of  ${}^6C_2$  matches  $\Rightarrow$  15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D, E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the teams that won both of their stage 2 matches are B, E and F.

 VIDEO SOLUTION



16. E

There are a total of  ${}^6C_2$  matches  $\Rightarrow$  15 matches. The first 9 matches are held in the first stage and remaining 6 in the second stage.

From the information given, we can conclude that the following matches were held in first stage:

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won)

One team won all matches. As B, C, D, E and F have lost at least one match each, A won all three matches. As A, B, D, E have won at least one match, C and F lost both matches.

From the matches already deduced, we can see that A needs to play 2 more matches, B two more matches and C and F one match each. As C and F lose all matches in stage 1, they cannot play against each other. F did not play against the leader i.e. A. Hence, the remaining matches are A-B (A won), A-C (A won), B-F (B won).

Thus, the stage 1 matches are

Stage 1: D-A (A won), D-C (D won), D-F (D won), E-B (B won), E-C (E won), E-F (E won), A-B (A won), A-C (A won), B-F (B won)

Thus Stage 2 matches are D-B, D-E, E-A, F-A, B-C and C-F (all matches - stage 1 matches)

As A lost both matches, F and E must have won the match vs A. As F won against A, F won both its matches and C lost both its matches. One more team lost both its matches. As B, E and F have won at least one match and A and C have been discussed previously, D must have lost both matches. Hence, stage 2 results are:

Stage 2: D-B (B won), D-E (E won), E-A (E won), F-A (F won), B-C (B won) and C-F (F won)

Hence, the wins by each team are A (3), B(4), C(0), D(2), E(4), F(2). Hence, D and F won exactly 2 matches.

 VIDEO SOLUTION

17. **D**

Germany and Argentina have 2 games each and Spain and Pakistan have won 1 game each. We also know that Germany beat Spain.

So Germany won its second game against one among SA/ NZ.

Similarly Argentina must have won one game against Pakistan and another against SA/ NZ.

Argentina scored 2 goals and conceded 0 goals. So it must have won both the games 1-0.

NZ conceded 6 goals. The only possibility is it lost 5-1 to Spain and lost 1-0 to Argentina.

As Spain conceded 2 goals it must have lost its opening game to Germany by 1-0.

Now, the results of all the games can be known.

Following are the possible match result after 2 rounds :

|     |     |     |
|-----|-----|-----|
| GER | 1-0 | SPA |
| GER | 2-1 | SA  |
| ARG | 1-0 | PAK |
| ARG | 1-0 | NZ  |
| SPA | 5-1 | NZ  |
| PAK | 2-0 | SA  |

HENCE OPTION D IS THE CORRECT ANSWER.

 VIEW SOLUTION

18. **B**

Following are the possible match result of 2 rounds :

|     |     |     |
|-----|-----|-----|
| GER | 1-0 | SPA |
| GER | 2-1 | SA  |
| ARG | 1-0 | PAK |
| ARG | 1-0 | NZ  |
| SPA | 5-1 | NZ  |
| PAK | 2-0 | SA  |

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 VIDEO SOLUTION

 VIDEO SOLUTION

24. C

If Elena Dementieva and Serena Williams lose in the second round, Nadia petrova and patty schnyder will go through.

Hence ,in the quarter finals, following seed no. players will be in quarter finals: 1,2,3,4,5,11,7,9.

So, now Maria Sharapova is ranked is 1 so she'll play 9th seed player who is Nadia Petrova.

Hence option C.

25. C

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**

|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

$$P1 - 88.6$$

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

$$P1 - 88.6$$

$$G - 89.6$$

$$B - 87.6$$



Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

Hence, P8 and P10 got the doubles.

 VIDEO SOLUTION



**CAT Formula PDF**



26. D

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**

|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

Hence, P1 won the silver medal.

 VIDEO SOLUTION

27. C

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**

|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

$$P1 - 88.6$$

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

$$P1 - 88.6$$

$$G - 89.6$$

$$B - 87.6$$

Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P7 improved by a total of 2.4.

 VIDEO SOLUTION

28. C

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**

|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

$$P1 - 88.6$$

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

$$P1 - 88.6$$

$$G - 89.6$$

$$B - 87.6$$

Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P8 comes between P9 and P6.

$$\text{Hence, } 82.5 < P8 < 84.1$$

$$P8 = 82.7$$

 VIDEO SOLUTION

29. A

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**



|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

P1 - 88.6

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

P1 - 88.6

G - 89.6

B - 87.6

Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P7 the gold winner had already received rank 1 at the end of Round 5. Hence, he was the last one to throw in the tournament.

 VIDEO SOLUTION

### 30.C

Let us arrange the players in the order in which they throw in each round.

**Round 1:** Here the players throw in order of their initial seeds so the order is as follows:

| Throw | Player |
|-------|--------|
| 1     | P1     |
| 2     | P2     |
| 3     | P3     |
| 4     | P4     |
| 5     | P5     |
| 6     | P6     |
| 7     | P7     |
| 8     | P8     |
| 9     | P9     |
| 10    | P10    |

So, their rank at the start of Round 2 is in order of their throws in the first round. Also, we need to consider the same order for people having invalid throws.

**Round 2:** P2, P4, P8, P10 had the same relative rankings since they all have invalid throws, that is, 0 metres. Rest are arranged as per their throw distances.

| Throw | Round 1 | Round 2 |
|-------|---------|---------|
| 1     | P1      | P7      |
| 2     | P2      | P5      |
| 3     | P3      | P9      |
| 4     | P4      | P1      |
| 5     | P5      | P6      |
| 6     | P6      | P3      |
| 7     | P7      | P2      |
| 8     | P8      | P4      |
| 9     | P9      | P8      |
| 10    | P10     | P10     |

Now, in round 2, only P8 and P10 had valid throws. Hence, their order will change at the start of Round 3, however, the remaining order stays the same. That is, P8 and P10 will move up in the table and occupy some higher places, whereas some of the others may move down consequently.

**Round 3:** In Round 3, we can see that P1 improved his score from 82.9 to 88.6. The other 2 participants did not improve their scores. Also after Round 3, P8 and P10 qualify, where one of P8 or P10 is at the sixth position. So at the end of Round 3, we can say that P6, P3, P2, and P4 are at the bottom 4 positions(ranks). One of P8 or P10 is at the sixth position.  $P1 > P7 > P5 > P9$ .

**So at the end of round 3, the ranks are as follows:**

|        |
|--------|
| P1     |
| P7     |
| P5     |
| P9     |
| P8/P10 |
| P6     |
| P3     |
| P2     |
| P4     |

The other person between P8/P10 can go anywhere between Rank 1 and Rank 5.

Now let us consider the two players who got a double. Doubles happen in the transition between rounds.

1 -> 2 - Not possible

2 -> 3 - Possible if P10 reaches Rank 1 after round 2.

3 -> 4 - P8/P10 who is the last among qualifying will be the first to throw. So, here it definitely happens.

4 -> 5 AND 5 -> 6 not possible.

So, after Round 2, definitely, P10 reaches the top of the ranking. P8 is at the bottom. Hence, after Round 3, P10 either retains rank 1 or P1 surpasses him and P10 becomes Rank 2.

So, two combinations are possible at the end of Round 3:

| Rank after Round 3 |        |
|--------------------|--------|
| Case 1             | Case 2 |
| P10                | P1     |
| P1                 | P10    |
| P7                 | P7     |
| P5                 | P5     |
| P9                 | P9     |
| P8                 | P8     |

Now, we know that in each of the rounds in phase 2, only one player improves his score. Also, P8 and P10 cannot win medals. Hence, in case 1, three of P1, P7, P5 and P9 will improve their scores by  $x$  and reach the top 3 positions. However, the top 3 positions' distances are in AP.

$$P1 - 88.6 + x$$

$$P7 - 87.2 + x$$

$$P5 - 86.4 + x$$

$$P9 - 84.1 + x$$

The differences do not satisfy the condition. Hence, case 1 is invalidated.

Case 2: Here, P1 definitely wins a medal, and P10 does not. So, two of P7, P5 and P9 jumps above P10. Now, if we have three different people increasing their scores or distances in each of the three rounds, again we would not get a difference of 1 among the Gold, Silver and Bronze medallists. Hence, one of them increases his score twice and the other increases his score twice and none of them is P1.

Let us take the cases where P1 is individually the G, S and B medallists.

Case 1: P1 is a G medallist.

$$P1 - 88.6$$

The silver medallist is 87.6 and the bronze medallist is 86.6 metres. However, P10 has thrown for a distance that is greater than 87.2 metres. Hence, in this case, he would be the B medallist. Hence, this is not the right case.

Case 2: P1 is the S medallist.

$$P1 - 88.6$$

$$G - 89.6$$

$$B - 87.6$$

Now, if we see the differences

$$89.6 - 87.2 = 2.4$$

$$87.6 - 86.4 = 1.2$$

This satisfies the condition that P7 has increased his score twice to become the gold winner and P5 has increased it once to become the bronze winner.

Hence, P1 - Silver

P7 - Gold

P5 - Bronze

P1 - 88.6 m.

 VIDEO SOLUTION



31. **D**

There are 28 matches in the 1st round of 8 player group so in total 28 wins. To find this, we need the top 5 teams to win nearly equal matches. So Let each of the top 5 teams win 5 matches each and the remaining 3 matches are won by the bottom 2 teams. The qualifiers will be decided based on the tie breaking rules. Hence, even with 5 wins a team may end up not qualifying for the next round. Thus, option D is the correct answer.

 VIDEO SOLUTION

32. **C**

In 1st stage there were 2 groups each group played 28 matches as  ${}^8C_2 = 28$ . So total 56 matches after 1st stage. IN 1st round of stage 2, 4 matches are played and in the next round we have 4 teams. The winner is one who defeats every team out of remaining 3 matches. so 3 matches more needed. In total  $56+4+3=63$ .

 VIDEO SOLUTION

33. **B**

Let us first try to find the maximum number of points with which a team cannot advance to the second stage.

For this to happen the top five teams should have highest possible equal number of points and one team does not advance after the tie breaker.

To maximize the points of top 5 teams all of the top 5 teams should win their matches with the bottom 3 teams. Now each team has 3 points.

The top 5 teams play  $5C_2 = 10$  matches among themselves. There are 10 points to be won from these 10 matches.

Now as all the top 5 teams should have equal number of points each team gets 2 points from these games.

In all each top 5 team gets  $3+2 = 5$  points each.

In this case after getting 5 points also one of the top 5 teams will not advance to the second round.

Thus the maximum number of points with which a team cannot advance to the second stage is 5.

Therefore at least 6 wins are required for a team to guarantee its advancement to the next stage.

 VIDEO SOLUTION

34. **C**

2nd stage 1st round has 8 teams , 2nd round will have 4 teams and 3rd round 2 teams where the winner wins .

[VIDEO SOLUTION](#)

35. C

Consider the below case.

| Group 1 |   |  | Group 2 |   |
|---------|---|--|---------|---|
| A       | 7 |  | P       | 7 |
| B       | 6 |  | Q       | 6 |
| C       | 5 |  | R       | 5 |
| D       | 2 |  | S       | 4 |
| E       | 2 |  | T       | 3 |
| F       | 2 |  | U       | 2 |
| G       | 2 |  | V       | 1 |
| H       | 2 |  | W       | 0 |

If D wins the tournament he would have 5 wins. But if P loses in second round, he would have 7 wins. Hence option A is incorrect.

We can see that T has more wins than D. Hence option B is incorrect.

Consider the below scenario:

| QF | Wins | SF | Wins | F | Wins | Total wins |
|----|------|----|------|---|------|------------|
| A  | 1    | A  | 1    | A | 1    | 3          |
| B  | 1    | B  | 1    | B | 0    | 2          |
| C  | 1    | C  | 0    |   |      | 1          |
| D  | 1    | D  | 0    |   |      | 1          |
| P  | 0    |    |      |   |      | 0          |
| Q  | 0    |    |      |   |      | 0          |
| R  | 0    |    |      |   |      | 0          |
| S  | 0    |    |      |   |      | 0          |

We can see that 2 teams have one win each in the second stage. Hence option D is incorrect.

Consider the following scenario.

| Group 1 |   |  | Group 2 |   |
|---------|---|--|---------|---|
| A       | 7 |  | P       | 5 |
| B       | 6 |  | Q       | 5 |
| C       | 5 |  | R       | 5 |
| D       | 2 |  | S       | 5 |
| E       | 2 |  | T       | 5 |
| F       | 2 |  | U       | 1 |
| G       | 2 |  | V       | 1 |
| H       | 2 |  | W       | 1 |

If D wins the tournament he would have 5 wins. But T would lose the tournament with 5 wins. Hence option C is correct.

[VIEW SOLUTION](#)

**CAT Percentile Predictor**



36. C

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be  $a$  each and no. of goals scored in 1st and 5th match be  $b$  and  $c$  respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for  $b$  and  $c$  will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for  $b$  and  $c$  will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

 VIDEO SOLUTION

37.D

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for b and c will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

 VIDEO SOLUTION

38. C

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:



| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be  $a$  each and no. of goals scored in 1st and 5th match be  $b$  and  $c$  respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for  $b$  and  $c$  will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for  $b$  and  $c$  will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

 VIDEO SOLUTION

39. A

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be a each and no. of goals scored in 1st and 5th match be b and c respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for b and c will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for b and c will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

 VIDEO SOLUTION

40. B

A total of 12 goals were scored in 8 matches and each player scored atleast one goal and no of goals scored by each one of them is distinct so the possible number of goals scored by the players can be (1,2,3,6) or (1,2,4,5).

From statement 4 we know that Bimal scored 4 goals and since Harita scored more goals than Bimal so we can say that Harita scored 5 goals and the only case possible for total goals scored by each of the players is (1,2,4,5).

Now using statement 1, statement 3 and statement 4 we can say that the three consecutive matches in which Bimal scored will be 5th, 6th and 7th matches as Harita scored in 4th and 8th matches and we get the following table:

| Matches | Goals Scored | Players  |
|---------|--------------|----------|
| Match-1 |              | Bimal-1  |
| Match-2 | 1            |          |
| Match-3 |              |          |
| Match-4 | 1            | Harita-1 |
| Match-5 |              | Bimal-1  |
| Match-6 | 1            | Bimal-1  |
| Match-7 |              | Bimal-1  |
| Match-8 | 1            | Harita-1 |

From statement 5 and 6, we can conclude that the highest number of goals were scored in Match 1.

Let the no. of goals scored in 3rd and 7th match be  $a$  each and no. of goals scored in 1st and 5th match be  $b$  and  $c$  respectively.

Therefore,  $2a+b+c=8$

If  $a=1$ , then  $b+c=6$  therefore possible solutions for  $b$  and  $c$  will be 2 and 4 only

If  $a=2$ , then  $b+c=4$  therefore possible solution for  $b$  and  $c$  will be 1 and 3 only but since highest goals scored is in Match 1 so then no. of goals scored in match 1 must be 3 and Harita must have scored 3 goals in match 1 as Harita scored 5 goals in exactly 3 matches. Therefore, we can see this is not possible because then the no. of goals scored in Match 1 becomes 4.

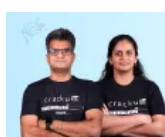
Therefore the only possible solution is  $a=1$ ,  $b=4$  and  $c=2$

| Matches | Goals Scored | Players           |
|---------|--------------|-------------------|
| Match-1 | 4            | Bimal-1, Harita-3 |
| Match-2 | 1            |                   |
| Match-3 | 1            |                   |
| Match-4 | 1            | Harita-1          |
| Match-5 | 2            | Bimal-1,          |
| Match-6 | 1            | Bimal-1           |
| Match-7 | 1            | Bimal-1           |
| Match-8 | 1            | Harita-1          |

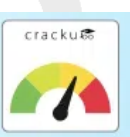
The remaining 3 goals were scored in the match 2, 3 and 5 by Amla and Sarita in some order.

| Matches | Goals Scored | Players                |
|---------|--------------|------------------------|
| Match-1 | 4            | Bimal-1, Harita-3      |
| Match-2 | 1            | Amla/Sarita-1          |
| Match-3 | 1            | Amla/Sarita-1          |
| Match-4 | 1            | Harita-1               |
| Match-5 | 2            | Bimal-1, Amla/Sarita-1 |
| Match-6 | 1            | Bimal-1                |
| Match-7 | 1            | Bimal-1                |
| Match-8 | 1            | Harita-1               |

 VIDEO SOLUTION



# CAT Score Calculator



41.1

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

Deepak got exactly one point in only R3.

Hence 1 is correct answer.

 VIDEO SOLUTION

42. 4

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:



|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

Arun bid high in R1,R2, R.x, R.z hence, 4 is correct answer.

 VIDEO SOLUTION

43. 4

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

Bikram bid Lo in R1,R2,R3,R.x. Hence 4 is correct answer.

 VIDEO SOLUTION

44.2

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

All the players made identical bids in R3 and R.z

 VIDEO SOLUTION

45. **A**

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

The bids by Arun, Bankim, Charu and Dipak, respectively in the first round are HLLH.

Hence Option A is correct.

VIDEO SOLUTION



46. A

Let 'H' represents Hi and 'L' represents Lo.

Given if they bid

Case 1: HHHH then all players gets -1 points.

Case 2: HHHL => H gets +1 and L gets -3.

Case 3: HHLL => H gets +2 and L gets -2.

Case 4: HLLL => H gets +3 and L gets -1.

Case 5: LLLL => every player gets +1.

From the given information we can draw the following table:

|   | R1 | R2 | R3 | T1 | R4 | R5 | R6 | T2 |
|---|----|----|----|----|----|----|----|----|
| A |    |    |    | 6  |    |    |    | 7  |
| B |    |    |    | -2 |    |    |    | -1 |
| C |    |    |    | -2 |    |    |    | -5 |
| D | D1 | D3 | D2 | 2  |    |    |    | -1 |

\*\*T1 is the cumulative of points till Round 3 and T2 is sum of points till round 6.

\*\*Arun, Bankim, Charu, and Dipak are represented by A, B, C, D respectively.

From point 3,  $D1 > D2 > D3$

D scored 2 points till round R3 and  $D1 > D3 > D2$  the possible scenarios are :

Case D1: 3,2,-3

In this case the points of A in R1, R3, R2 will be -1,2/-2, 1 in any possible combination the sum will not be 6. So, this case is invalid.

Case D2: 2,1,-1

In this case the points of A in R1, R3, R2 will be 2/-2, 1/-3, -1/3 so, if the points in R1, R3, R2 are 2,1,3 the case is valid and no other cases are possible.

Case D3: 3,1,-2

In this case the points of A in R1, R3, R2 will be -1, 1/-3/1, 2/-2 in any possible combination the sum will not be 6. So, this case is invalid.

∴ Points of A,D in (R1,R2,R3) are (2,3,1) and (2,-1,1) respectively.

Since A got +3 in R2, he is only the one to bid h in R2 and points of B and C in round 2 are -1,-1 i.e they bid L, L.

Since A and D got 2 points each in R1, C and B must have got -2, -2 i.e they bid L, L.

Since A and D got 1 point in R3, C and B must also have got 1 in R3 i.e they bid L, L.

With this data, the table now looks like:

|   | R1      | R2      | R3     | T1 | R4 | R5 | R6 | T2 |
|---|---------|---------|--------|----|----|----|----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  |    |    |    | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 |    |    |    | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  |    |    |    | -1 |

No information is given about the individual scores in R4, R5, R6.

Given In exactly two out of the six rounds, Arun was the only player who bid Hi.

Let R.x, R.y, R.z represent R4, R5, R6 in any order.

Let A bid H in R.x=> B,C,D bid L.

The table now looks like:

|   | R1      | R2      | R3     | T1 | R.x     | R.y | R.z | T2 |
|---|---------|---------|--------|----|---------|-----|-----|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  |     |     | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L |     |     | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L |     |     | -1 |

For A,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=-2$

For B,  $R.x+R.y+R.z=1 \Rightarrow R.y+R.z=2$

For C,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

For D,  $R.x+R.y+R.z=-3 \Rightarrow R.y+R.z=-2$

(R.y, R.z) for A can be (-3,1) or (-1,-1)

Case A1:

If for A, (R.y, R.z)=(-3,1)

Since for both C,D:  $R.y+R.z=-2$

We can't get any combination such that the total points of B,C,D are obtained.

Case A2:

If for A, (R.y, R.z)=(-1,-1).the (R.y, R.z) of B,C,D can be (3,-1), (-1,-1), (-1,-1) and they must have bid (H,H), (L,H), (L,H) respectively while A must have bid (L, H)

Hence this case is valid.

The final table looks like:



|   | R1      | R2      | R3     | T1 | R.x     | R.y     | R.z     | T2 |
|---|---------|---------|--------|----|---------|---------|---------|----|
| A | 2<br>H  | 3<br>H  | 1<br>L | 6  | 3<br>H  | -1<br>L | -1<br>H | 7  |
| B | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | 3<br>H  | -1<br>H | -1 |
| C | -2<br>L | -1<br>L | 1<br>L | -2 | -1<br>L | -1<br>L | -1<br>H | -5 |
| D | 2<br>H  | -1<br>L | 1<br>L | 2  | -1<br>L | -1<br>L | -1<br>H | -1 |

R2 is correct answer.

 VIDEO SOLUTION

47. **B**

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

| Group - A     | Group - B            |
|---------------|----------------------|
| Aruna vs Biju |                      |
| Azul vs Aruna | Biju vs Brinda/ Brij |
| Azul vs Arif  | Biju vs Brinda/ Brij |

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

| Group - C        | Group - D          |
|------------------|--------------------|
| Chitra           | vs Dipen           |
| Chetan vs Chitra | Dipen vs Deb/Donna |
| Chetan vs Chavvi | Dipen vs Deb/Donna |

Aruna plays Chitra in the finals and wins the final round.

Aruna is the winner.

▶ VIDEO SOLUTION

48. D

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

| Group - A     | Group - B            |
|---------------|----------------------|
| Aruna         | vs Biju              |
| Azul vs Aruna | Biju vs Brinda/ Brij |
| Azul vs Arif  | Biju vs Brinda/ Brij |

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

| Group - C        | Group - D          |
|------------------|--------------------|
| Chitra           | vs Dipen           |
| Chetan vs Chitra | Dipen vs Deb/Donna |
| Chetan vs Chavvi | Dipen vs Deb/Donna |

Aruna plays Chitra in the finals and wins the final round.

Chitra and Dipen played in the semifinals

VIDEO SOLUTION

49. D

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

| Group - A     | Group - B            |
|---------------|----------------------|
| Aruna         | vs Biju              |
| Azul vs Aruna | Biju vs Brinda/ Brij |
| Azul vs Arif  | Biju vs Brinda/ Brij |

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

| Group - C        | Group - D          |
|------------------|--------------------|
| Chitra           | vs Dipen           |
| Chetan vs Chitra | Dipen vs Deb/Donna |
| Chetan vs Chavvi | Dipen vs Deb/Donna |

Aruna plays Chitra in the finals and wins the final round.

Brij was the player from group B who played Chitra. Aruna played in finals, Chetan in round 2, and Dipen in semi finals

▶ VIDEO SOLUTION

50. A

Group A :

Since Aruna played 3 games if she belongs to rank 2 or rank 3 in her group she must have reached the semifinals and lost in the semifinals. But for this case, she must play against rank 1 and rank 3 in her group. But she did not play against Arif from her group.

Hence Aruna was ranked 1 in her group, Among Azul and Arif one of them was ranked 2 and the other was ranked 3. Azul defeated Arif in the first round and in the second round lost to Aruna. Aruna played her first round with Azul and won the round and played against the winner from group B and defeated them and moved to the finals.

Group B :

Brij did not play against Brinda. Biju played three games, Brij and Brinda played one game each.

Since Brij and Brinda played only one game each one of them was ranked 1 and the other was ranked 2 and 3. Brij did not play against Brinda. We are aware that Aruna reached finals and hence the person from Group B did not reach the finals. Biju played three games and hence must have played with Brij/Brinda in the first round and won the round. Plays with Brij/ Brinda and wins the second round. Plays with Aruna and loses the third round.

| Group - A     | Group - B            |
|---------------|----------------------|
| Aruna         | vs Biju              |
| Azul vs Aruna | Biju vs Brinda/ Brij |
| Azul vs Arif  | Biju vs Brinda/ Brij |

Group - C :

Chitra played 2 matches and Chetan played 2 matches. For Chitra to play 2 matches if she is rank 2 or rank 3 in her group. She must at least reach the semifinals. But in this case, Chetan will be defeated in his first round. So Chitra must be ranked 1 in her group and Chetan must be ranked 2 or rank 3 in his group. He defeats Chhavi in his first round and loses to Chitra in his second round. Chitra plays Chetan in her first round, wins over the winner of group D in her second round, and loses in the final against Aruna as per condition 1.

Group -D :

The person from Group D did not reach the finals because Chitra reached the finals. In order for Dipen to play 3 matches before his finals. Dipen must be ranked 2 or rank 3 in his group and plays Deb and Donna in the first two rounds in any order and wins over both of them. Dipen loses to Chitra in his third round.

| Group - C        | Group - D          |
|------------------|--------------------|
| Chitra           | vs Dipen           |
| Chetan vs Chitra | Dipen vs Deb/Donna |
| Chetan vs Chavvi | Dipen vs Deb/Donna |

Aruna plays Chitra in the finals and wins the final round.

Dipen was ranked 2 or 3 in his group

[VIDEO SOLUTION](#)



51. A

The ranking list would be in the order A, B, C, D....., X, Y, Z. Now the A wins all his 25 matches, B wins 24 matches and lost to A. C wins 23 matches and lost to A and B. In this way N 12 matches and loses 13 matches to A,B,C,D...,M

[VIEW SOLUTION](#)

52. A

| Round no. | Matches | Winners      |
|-----------|---------|--------------|
| 1         | 58      | 58           |
| 2         | 29      | 29 or (28+1) |
| 3         | 14      | (14+1)       |
| 4         | 7       | (7+1 = 8)    |
| 5         | 4       | 4            |
| 6         | 2       | 2            |
| 7         | 1       | 1            |
| Total     | 115     |              |

In second round, number of winners were odd hence 1 winner will be played in next round. Since in third round also number of winners are odd, then last round player will be played in this round and one another player will be played in next round.

Now in 4th round total winners will be (7+1), this 1 winner belongs to 3rd round where he was sent to play in next round.

Summation of all matches will be 115.

[VIEW SOLUTION](#)

53. A

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 |       |       |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       | 11    |        | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> | 12    | 11    | 5      | 12     |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

 VIDEO SOLUTION

54.6

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table



|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 |       |       |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       | 11    |        | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> | 12    | 11    | 5      | 12     |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

 VIDEO SOLUTION

55.6

Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 |       |       |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       | 11    |        | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 | 11    | 11    | 8      | 10     |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 | 12    | 12    | 4      | 12     |
| Round 8 | 13    | 11    | 6      | 10     |

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 | 11    | 11    | 8      | 10     |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 | 12    | 11    | 5      | 12     |
| Round 7 | 12    | 12    | 4      | 12     |
| Round 8 | 13    | 11    | 6      | 10     |

The Games which had a bet of Rs.2 are as following

Round 1 - Palak vs Qasim

Round 2 - Qasim vs Suresh

Round 3 - Palak vs Suresh

Round 4 - Ritesh vs Suresh

Round 5 - None

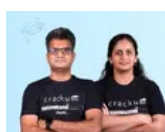
Round 6 - Palak vs Ritesh

Round 7 - None

Round 8 - Ritesh vs Suresh

Hence total number of games that were played with a bet of 2 is 6

VIDEO SOLUTION



## CAT Daily Targets

Step by step, heading towards 99+ percentile




Its given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount,i.e., 40 in each round remains the same. So we get the following table

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       |       |        |        |
| <b>Round 4</b> |       |       |        |        |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       |       |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       |       |        |        |
| <b>Round 4</b> | 11    | 11    |        |        |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       |       |        |        |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> |       | 11    |        | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    |       |        | 10     |

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.



|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 | 11    | 11    | 8      | 10     |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 | 12    | 11    | 5      | 12     |
| Round 7 | 12    | 12    | 4      | 12     |
| Round 8 | 13    | 11    | 6      | 10     |

For round 6 the pairs formed were Pulak-Ritesh and Qasim-Suresh. For round 4 the pairs formed were Pulak-Qasim and Ritesh-Suresh.

Therefore the pairs formed for round 5 were Pulak-Suresh and Qasim-Ritesh

 VIDEO SOLUTION

57.D

It's given that the table shows every time a player had ₹10 at the end of a round, as well as every time, at the end of a round, a player had either the minimum or the maximum amount that he would have had across the eight rounds, which means that in Pulak's column the numbers possible are 11 or 12 only, similarly in Qasim's column numbers possible are only 9 or 11 only, similarly in Ritesh's column numbers possible are 5,6,7,8 or 9 only and in Suresh's column numbers possible are 9,11 or 12 only.

Everyone started with 10 rupees and the total amount, i.e., 40 in each round remains the same. So we get the following table

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 |       |       |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       |       |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

We are given that at the end of 4th round both Pulak and Qasim had the same amount with them and that amount possible is only 11.

As per the information given in the set the only possible amount with Qasim at the end of round 6 is 11.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    |        |        |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amounts with Pulak and Qasim at the end of round 5 is decreased by 1 from the amounts which they had at the end of round 4, since the total amount will be same at the end of every round, so amounts with Ritesh and Suresh at the end of round 5 is increased by 1 from the amounts which they had at the end of round 4.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       |       |        |        |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Amount with Suresh at the end of round 2 is 8 and at the end of round 4 is 12, therefore, amount with Suresh at the end of round 3 is 10.

Amount with Qasim at the end of round 3 is either 9 or 11. If the amount is 9 then either amount with Ritesh has to be 10 or amount with Pulak has to be 14, both of which is not possible. Therefore, amount with Qasim at the end of round 3 is 11.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 |       | 11    |        | 10     |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 |       | 11    |        |        |
| Round 7 |       | 12    | 4      |        |
| Round 8 | 13    |       |        | 10     |

Now Qasim has the same amount at the end of round 3 and round 4 so there must be another person whose amount is also same at the end of round 3 and round 4. This person can only be Pulak. So at the end of round 3 Pulak has 11 rupees and Ritesh has 8 rupees.

The only possible amount with Qasim at the end of round 8 can be 11 and therefore the amount at the end of round 8 with Ritesh will be 6.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> |       | 12    | 4      |        |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Ritesh at the end of round 8 is 2 more than the amount with him at the end of round 7, so the amount with either Pulak or Suresh at the end of round 7 has to be 2 more than the amount at the end of round 8. This is only possible for Suresh who must have 12 rupees at the end of round 7 as Pulak cannot have rupees 15 at the end of round 7.

|                | Pulak | Qasim | Ritesh | Suresh |
|----------------|-------|-------|--------|--------|
| <b>Round 1</b> | 12    | 8     | 10     | 10     |
| <b>Round 2</b> | 13    | 10    | 9      | 8      |
| <b>Round 3</b> | 11    | 11    | 8      | 10     |
| <b>Round 4</b> | 11    | 11    | 6      | 12     |
| <b>Round 5</b> | 10    | 10    | 7      | 13     |
| <b>Round 6</b> |       | 11    |        |        |
| <b>Round 7</b> | 12    | 12    | 4      | 12     |
| <b>Round 8</b> | 13    | 11    | 6      | 10     |

The amount with Qasim at the end of round 7 is 1 more than the amount with him at the end of round 6. So there must be another person whose amount at the end of round 7 is 1 less than the amount him at the end of round 6. This only possible person can be Ritesh. So amount with Ritesh at the end of round 6 will be 5.

If the amount with Suresh at the end of round 6 is 11 then the amount with Pulak at the end of round 6 will be 13 which is not possible. Therefore the amount with Suresh at the end of round 6 is 12 and the amount with Pulak at the end of round 6 is 12.

|         | Pulak | Qasim | Ritesh | Suresh |
|---------|-------|-------|--------|--------|
| Round 1 | 12    | 8     | 10     | 10     |
| Round 2 | 13    | 10    | 9      | 8      |
| Round 3 | 11    | 11    | 8      | 10     |
| Round 4 | 11    | 11    | 6      | 12     |
| Round 5 | 10    | 10    | 7      | 13     |
| Round 6 | 12    | 11    | 5      | 12     |
| Round 7 | 12    | 12    | 4      | 12     |
| Round 8 | 13    | 11    | 6      | 10     |

VIDEO SOLUTION

58.D

From the condition given in the premise, we can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A |   | X | X | X | X |   | 8  |   |   |   |    | 18 |
| B |   |   | X | X | X | X | 2  |   |   |   |    | 5  |
| C |   |   |   | X | X | X | 3  | X |   |   |    | 6  |
| D |   |   |   |   | X | X | 6  | X | X |   |    | 6  |
| E |   |   |   |   |   | X | 3  | X | X | X |    | 10 |
| F |   |   |   |   |   |   | 10 | X | X | X | X  | 10 |
| G | X |   |   |   |   |   | 17 |   | X | X | X  | 17 |
| H | X | X |   |   |   |   | 1  |   |   | X | X  | 4  |
| I | X | X | X |   |   |   | 2  |   |   |   | X  | 17 |
| J | X | X | X | X |   |   | 14 |   |   |   |    | 17 |

The information known about Rounds 1 through 6:

1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.
2. Amita (A) scored 1,7 points then she scored 7 in the first round.
3. Bala (B) scored 1 point in both the rounds 1st and 2nd.
4. Ikea (I) scored 1 point in the round 4th and 5th.
5. Gordon(G- 7,7,3 ) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  |   |   |   |    | 18 |
| B | 1 | 1 | X | X | X | X | 2  |   |   |   |    | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X |   |   |    | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X |   |    | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X |    | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 |   | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  |   |   | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  |   |   |   | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 |   |   |   |    | 17 |

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.
2. Ikea scored 15 points (1,7,7) in three rounds respectively.
3. Eric scored 7 in round 10.
4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  | 7 | 0 | 0 | 3  | 18 |
| B | 1 | 1 | X | X | X | X | 2  | 0 | 0 | 3 | 0  | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X | 1 | 1 | 1  | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X | 0 | 0  | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X | 7  | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 | 0 | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  | 0 | 3 | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  | 1 | 7 | 7 | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 | 3 | 0 | 0 | 0  | 17 |

Hence option D is correct.

[VIDEO SOLUTION](#)

59. D

From the condition given in the premise, we can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A |   | X | X | X | X |   | 8  |   |   |   |    | 18 |
| B |   |   | X | X | X | X | 2  |   |   |   |    | 5  |
| C |   |   |   | X | X | X | 3  | X |   |   |    | 6  |
| D |   |   |   |   | X | X | 6  | X | X |   |    | 6  |
| E |   |   |   |   |   | X | 3  | X | X | X |    | 10 |
| F |   |   |   |   |   |   | 10 | X | X | X | X  | 10 |
| G | X |   |   |   |   |   | 17 |   | X | X | X  | 17 |
| H | X | X |   |   |   |   | 1  |   |   | X | X  | 4  |
| I | X | X | X |   |   |   | 2  |   |   |   | X  | 17 |
| J | X | X | X | X |   |   | 14 |   |   |   |    | 17 |

The information known about Rounds 1 through 6:

1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.
2. Amita (A) scored 1,7 points then she scored 7 in the first round.
3. Bala (B) scored 1 point in both the rounds 1st and 2nd.
4. Ikea (I) scored 1 point in the round 4th and 5th.
5. Gordon(G- 7,7,3 ) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  |   |   |   |    | 18 |
| B | 1 | 1 | X | X | X | X | 2  |   |   |   |    | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X |   |   |    | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X |   |    | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X |    | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 |   | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  |   |   | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  |   |   |   | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 |   |   |   |    | 17 |

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.
2. Ikea scored 15 points (1,7,7) in three rounds respectively.
3. Eric scored 7 in round 10.
4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  | 7 | 0 | 0 | 3  | 18 |
| B | 1 | 1 | X | X | X | X | 2  | 0 | 0 | 3 | 0  | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X | 1 | 1 | 1  | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X | 0 | 0  | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X | 7  | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 | 0 | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  | 0 | 3 | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  | 1 | 7 | 7 | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 | 3 | 0 | 0 | 0  | 17 |

Hence option D is correct.

 VIDEO SOLUTION

60. B

From the condition given in the premise, we can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A |   | X | X | X | X |   | 8  |   |   |   |    | 18 |
| B |   |   | X | X | X | X | 2  |   |   |   |    | 5  |
| C |   |   |   | X | X | X | 3  | X |   |   |    | 6  |
| D |   |   |   |   | X | X | 6  | X | X |   |    | 6  |
| E |   |   |   |   |   | X | 3  | X | X | X |    | 10 |
| F |   |   |   |   |   |   | 10 | X | X | X | X  | 10 |
| G | X |   |   |   |   |   | 17 |   | X | X | X  | 17 |
| H | X | X |   |   |   |   | 1  |   |   | X | X  | 4  |
| I | X | X | X |   |   |   | 2  |   |   |   | X  | 17 |
| J | X | X | X | X |   |   | 14 |   |   |   |    | 17 |

The information known about Rounds 1 through 6:

1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.
2. Amita (A) scored 1,7 points then she scored 7 in the first round.
3. Bala (B) scored 1 point in both the rounds 1st and 2nd.
4. Ikea (I) scored 1 point in the round 4th and 5th.

5. Gordon(G- 7,7,3 ) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  |   |   |   |    | 18 |
| B | 1 | 1 | X | X | X | X | 2  |   |   |   |    | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X |   |   |    | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X |   |    | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X |    | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 |   | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  |   |   | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  |   |   |   | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 |   |   |   |    | 17 |

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.

2. Ikea scored 15 points (1,7,7) in three rounds respectively.

3. Eric scored 7 in round 10.

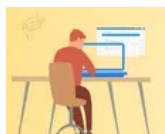
4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  | 7 | 0 | 0 | 3  | 18 |
| B | 1 | 1 | X | X | X | X | 2  | 0 | 0 | 3 | 0  | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X | 1 | 1 | 1  | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X | 0 | 0  | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X | 7  | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 | 0 | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  | 0 | 3 | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  | 1 | 7 | 7 | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 | 3 | 0 | 0 | 0  | 17 |

Hence option B is correct.

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## CAT Sectional Tests

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61. D

From the condition given in the premise, we can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A |   | X | X | X | X |   | 8  |   |   |   |    | 18 |
| B |   |   | X | X | X | X | 2  |   |   |   |    | 5  |
| C |   |   |   | X | X | X | 3  | X |   |   |    | 6  |
| D |   |   |   |   | X | X | 6  | X | X |   |    | 6  |
| E |   |   |   |   |   | X | 3  | X | X | X |    | 10 |
| F |   |   |   |   |   |   | 10 | X | X | X | X  | 10 |
| G | X |   |   |   |   |   | 17 |   | X | X | X  | 17 |
| H | X | X |   |   |   |   | 1  |   |   | X | X  | 4  |
| I | X | X | X |   |   |   | 2  |   |   |   | X  | 17 |
| J | X | X | X | X |   |   | 14 |   |   |   |    | 17 |



The information known about Rounds 1 through 6:

1. Gordon(G) did not score consecutively in any two rounds.
2. Eric(E) and Fatima(F) both scored in a round.

By observing the table:

1. Jordan(J) scored 7 points in both the rounds 5th & 6th.
2. Amita (A) scored 1,7 points then she scored 7 in the first round.
3. Bala (B) scored 1 point in both the rounds 1st and 2nd.
4. Ikea (I) scored 1 point in the round 4th and 5th.
5. Gordon(G- 7,7,3 ) did not score consecutively in any two rounds so it scored in 2nd, 4th and 6th rounds respectively.

We can make the following table from the details given in the question.

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  |   |   |   |    | 18 |
| B | 1 | 1 | X | X | X | X | 2  |   |   |   |    | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X |   |   |    | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X |   |    | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X |    | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 |   | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  |   |   | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  |   |   |   | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 |   |   |   |    | 17 |

T: Total after the sixth round and TT: Total after the 10th round.

1. Only two players scored in three consecutive rounds. One of them was Chen. So He scored 1 point in the rounds 8th, 9th and 10th.
2. Ikea scored 15 points (1,7,7) in three rounds respectively.
3. Eric scored 7 in round 10.
4. Amita will score 3 in round 10, and 7 in round 7.

We can make the following table:

|   | 1 | 2 | 3 | 4 | 5 | 6 | T  | 7 | 8 | 9 | 10 | TT |
|---|---|---|---|---|---|---|----|---|---|---|----|----|
| A | 7 | X | X | X | X | 1 | 8  | 7 | 0 | 0 | 3  | 18 |
| B | 1 | 1 | X | X | X | X | 2  | 0 | 0 | 3 | 0  | 5  |
| C | 3 | 0 | 0 | X | X | X | 3  | X | 1 | 1 | 1  | 6  |
| D | 0 | 3 | 0 | 3 | X | X | 6  | X | X | 0 | 0  | 6  |
| E | 0 | 0 | 3 | 0 | 0 | X | 3  | X | X | X | 7  | 10 |
| F | 0 | 0 | 7 | 0 | 3 | 0 | 10 | X | X | X | X  | 10 |
| G | X | 7 | 0 | 7 | 0 | 3 | 17 | 0 | X | X | X  | 17 |
| H | X | X | 1 | 0 | 0 | 0 | 1  | 0 | 3 | X | X  | 4  |
| I | X | X | X | 1 | 1 | 0 | 2  | 1 | 7 | 7 | X  | 17 |
| J | X | X | X | X | 7 | 7 | 14 | 3 | 0 | 0 | 0  | 17 |

Hence option D is correct.

 VIDEO SOLUTION